A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Thermodynamics

This topic is about what heating actually does to matter. You will see where the energy goes when something warms up or melts, why there is a coldest possible temperature, and how the steady pressure of a gas comes from countless tiny collisions. By the end you will be able to derive the link between how fast molecules move and how hot the gas is.

Internal energy and temperature

Everything is made of molecules that are always moving and always pulling or pushing on their neighbours. So every molecule has two kinds of energy: **kinetic energy**, because it moves, and **potential energy**, because of the forces between it and the molecules around it (a bit like the energy stored in a stretched or squashed spring). Add these up over every molecule and you get the **internal energy**.

DEFINITION

Internal energy

The internal energy of a body is the sum of the **randomly distributed** kinetic and potential energies of all of its molecules.

The word **random** matters. At any instant some molecules are fast and some are slow, some are squeezed close to a neighbour and some are far away. You can't say what one molecule is doing, only what the total and the average are.

- **Solid:** molecules vibrate about fixed positions, held in place by strong bonds.
- **Liquid:** molecules are still close together, but they can slide past one another.
- **Gas:** molecules are far apart and move fast in random directions. The forces between them only matter during collisions.

Heating something increases its internal energy, and it can do that in two different ways:

- it can make the molecules move faster on average, raising their **kinetic** energy. You see this as a **rise in temperature**;
- it can pull the molecules further apart against the forces between them, raising their **potential** energy. This is what happens during **melting and boiling**, and the temperature stays **constant** while it happens.

KEY IDEA

Temperature is not internal energy

Temperature is a measure of the **average kinetic energy** of the molecules. Internal energy is the **total** energy of all of them. An iceberg at 0 °C has far more internal energy than a cup of tea at 80 °C, simply because it has vastly more molecules. Energy flows by heating from a higher **temperature** to a lower one, so heat flows from the tea to the iceberg, not the other way.

Specific heat capacity: ΔE = mcΔθ

Think about boiling water in a kettle. With twice as much water you need twice as much energy, because there are twice as many molecules to speed up. To warm the water by 20 K instead of 10 K you also need twice as much energy. So the energy is proportional to both the **mass** and the **temperature change**. The constant of proportionality tells you how much energy a particular material soaks up per kilogram per kelvin, and we call it the specific heat capacity, c.

Heating with no change of state

FORMULA SHEET

$$\Delta E = mc\,\Delta\theta$$

ΔE: energy transferred / J m: mass / kg c: specific heat capacity / J kg⁻¹ K⁻¹
$\Delta\theta$: temperature change / K or °C

DEFINITION

Specific heat capacity

The specific heat capacity of a substance is the energy needed to raise the temperature of **1 kg** of it by **1 K** (or 1 °C), without any change of state.

Water has an unusually large value, $c = 4180\ \text{J kg}^{-1}\,\text{K}^{-1}$. Aluminium is about 900 and copper about 390. That is why the sea warms and cools so slowly, and why a metal pan heats up much faster than the water inside it.

EXAM TIP

A temperature **change** of 1 K is exactly the same size as a change of 1 °C, so $\Delta\theta$ can be worked out in either. A temperature itself (like T in $pV = NkT$) must always be in kelvin.

Measuring c with an electric heater

Put an electric heater into a block of the material (or into some water) and measure the voltage V, current I and time t. The energy supplied is $\Delta E = VIt$, so $c = VIt/m\Delta\theta$. If you plot θ against t, the gradient of the straight part is P/mc. Heat lost to the surroundings makes $\Delta\theta$ too small, so c comes out **too large**. Insulating the block, and starting a few degrees below room temperature so that you finish the same amount above it, both reduce this error.

WORKED EXAMPLE

A 2.2 kW kettle contains 0.80 kg of water at 18 °C. (a) Calculate the minimum time it takes to bring the water to 100 °C. (b) In practice it takes 150 s. Calculate the efficiency of the kettle. ($c_{\text{water}} = 4180\ \text{J kg}^{-1}\,\text{K}^{-1}$)

Show solution

(a) Energy needed by the water:

$$\Delta E = mc\Delta\theta = 0.80 \times 4180 \times (100 - 18) = 2.74 \times 10^5\ \text{J}$$

The minimum time assumes every joule goes into the water:

$$t = \frac{\Delta E}{P} = \frac{2.742 \times 10^5}{2200} = 125\ \text{s}$$

(b) Energy actually supplied $= 2200 \times 150 = 3.30 \times 10^5$ J, so

$$\text{efficiency} = \frac{2.742 \times 10^5}{3.30 \times 10^5} = 0.83 = 83\%$$

The rest heated the kettle itself and escaped to the room.

Specific latent heat: ΔE = LΔm

Put a thermometer in melting ice and keep heating. The reading sits at 0 °C until the last bit of ice has gone. The energy hasn't disappeared: it is doing work against the forces holding the molecules together, so it increases their **potential** energy instead of their kinetic energy. Because temperature reflects the average kinetic energy, the temperature doesn't change. The word "latent" means hidden, because the thermometer can't see this energy.

The energy needed is proportional to how much substance changes state, which gives another simple equation:

Change of state at constant temperature

FORMULA SHEET

$$\Delta E = L\,\Delta m$$

ΔE: energy transferred / J L: specific latent heat / J kg^{-1} Δm: mass that changes state / kg

DEFINITION

Specific latent heat

The specific latent heat of a substance is the energy needed to change the state of **1 kg** of it **without any change in temperature**. It is called the latent heat of **fusion** for melting (solid ↔ liquid) and of **vaporisation** for boiling (liquid ↔ gas).

For water, $L_\text{f} = 3.34 \times 10^5$ J kg^{-1} and $L_\text{v} = 2.26 \times 10^6$ J kg^{-1}, almost seven times bigger. Melting only has to loosen the structure enough for the molecules to slide past each other. Boiling has to pull every molecule completely away from all of its neighbours, and the escaping vapour also has to push back the atmosphere as it expands.

Part of the heating curve	Equation	What sets its shape (constant power P)
ice warming	$\Delta E = mc_\text{ice}\Delta\theta$	gradient $\dfrac{\Delta\theta}{\Delta t} = \dfrac{P}{mc_\text{ice}}$
melting at 0 °C	$\Delta E = mL_\text{f}$	length $t = \dfrac{mL_\text{f}}{P}$
water warming	$\Delta E = mc_\text{water}\Delta\theta$	gradient $\dfrac{P}{mc_\text{water}}$, half as steep as for ice because $c_\text{water} \approx 2c_\text{ice}$
boiling at 100 °C	$\Delta E = mL_\text{v}$	length $t = \dfrac{mL_\text{v}}{P}$, the longest part by far

COMMON MISTAKE

Don't use $\Delta E = mc\Delta\theta$ across a change of state. Split the process into stages (warm, change state, warm again), find the energy for each stage with the right equation and the right value of c, then add them up.

WORKED EXAMPLE

0.200 kg of ice at −15 °C is heated until it becomes water at 40 °C. (a) Calculate the energy needed. (b) How long does this take with a 500 W heater, assuming no losses? ($c_{\text{ice}} = 2100\ \text{J kg}^{-1}\,\text{K}^{-1}$, $c_{\text{water}} = 4180\ \text{J kg}^{-1}\,\text{K}^{-1}$, $L_{\text{f}} = 3.34 \times 10^{5}\ \text{J kg}^{-1}$)

Show solution

Warm the ice from −15 °C to 0 °C:

$$\Delta E_1 = 0.200 \times 2100 \times 15 = 6300\ \text{J}$$

Melt it at 0 °C:

$$\Delta E_2 = 0.200 \times 3.34 \times 10^{5} = 66\,800\ \text{J}$$

Warm the water from 0 °C to 40 °C:

$$\Delta E_3 = 0.200 \times 4180 \times 40 = 33\,440\ \text{J}$$

(a) Total: $\Delta E = 6300 + 66\,800 + 33\,440 = 1.07 \times 10^{5}\ \text{J}$

(b) $t = \dfrac{\Delta E}{P} = \dfrac{1.065 \times 10^{5}}{500} = 213\ \text{s}$. Notice that melting takes more energy than the other two stages put together.

Core Practical 13: specific latent heat

CORE PRACTICAL

Determine the specific latent heat of vaporisation of water

Stand a beaker of water on a top-pan balance and put an immersion heater in it, connected through a joulemeter (or an ammeter, a voltmeter and a stopwatch, so that $E = VIt$). Bring the water to the boil and let it boil steadily. Then record the balance reading, start the clock, and record the reading again after a measured time (about 5 minutes).

The energy supplied boils off a mass Δm, so $L = \dfrac{E}{\Delta m} = \dfrac{Pt}{\Delta m}$.

A single run always gives too large a value, because some of the energy heats the room instead of boiling water. The better method is to repeat at several different powers and plot **power P against the rate of mass loss** $\Delta m/\Delta t$. If heat is lost at a steady rate P_{loss}:

$$P = L\,\frac{\Delta m}{\Delta t} + P_{\text{loss}}$$

This is $y = mx + c$: the **gradient is** L and the intercept on the power axis is the rate of heat loss, which the gradient method removes.

- Only start timing once the water is boiling steadily, otherwise some energy is still raising its temperature.
- Insulate the sides of the beaker to cut heat loss. Keep the heater fully covered by water.
- Splashing makes Δm too big, and steam condensing on the heater and dripping back makes it too small.
- **Safety:** steam and boiling water scald. Switch the heater on only when it is immersed, keep the connections dry, and let everything cool before you move it.

For the latent heat of **fusion** you can use a heater in a funnel of crushed ice and collect the meltwater. A second, identical funnel with no heater shows how much ice melts just from the warmth of the room, and you subtract that mass.

Core Practical 12: a thermistor thermostat

A **negative temperature coefficient (NTC) thermistor** is a semiconductor whose resistance **falls** as it warms up, because extra thermal energy frees more charge carriers. Put it in series with a fixed resistor R across a supply V_{in} and you have a potential divider. The p.d. across the fixed resistor is

$$V_{\text{out}} = V_{\text{in}} \frac{R}{R + R_{\text{th}}}$$

As the temperature rises, R_{th} falls, so the fixed resistor takes a bigger share of the supply and V_{out} **rises**. Connect V_{out} to a switching circuit (a transistor or relay) that turns a heater off above a set voltage, and you have a **thermostat**. To set it to a particular temperature you first need to know which voltage goes with which temperature. Finding that out is called **calibration**.

CORE PRACTICAL

Calibrate a thermistor in a potential divider circuit as a thermostat

Put the thermistor (waterproofed, or inside a test tube) in a beaker of water next to a thermometer. Start with ice and water near 0 °C and warm it slowly. At each temperature, stir, wait until the readings are steady, and record θ and V_{out}. Plot a **calibration curve** of V_{out} against θ.

To use the circuit as a thermostat, read the voltage that goes with your target temperature off the curve and set that as the switching voltage. The circuit is most sensitive (biggest change in voltage per kelvin) where the curve is steepest, which is around the temperature at which $R_{\text{th}} = R$. So choose a fixed resistor about equal to the thermistor's resistance at the temperature you care about.

- The thermistor and thermometer must be at the same temperature: stir, keep them close together and wait before each reading.
- Keep the current small so the thermistor doesn't heat itself.
- Read the thermometer at eye level to avoid parallax.

- **Safety:** use a low-voltage supply near water, and take care with hot water.

Absolute zero and the kelvin scale

Seal some air in a flask so its volume can't change, and heat it. The pressure rises in a straight line with temperature. Now imagine cooling it instead and follow the line down. Whatever gas you use, and however much of it, the line reaches **zero pressure at about −273 °C**.

Zero pressure means the molecules have stopped hitting the walls: they have the least kinetic energy they can possibly have. You can't cool anything below this, because you can't take away kinetic energy that isn't there. This temperature is called **absolute zero**, and the kelvin scale starts from it.

DEFINITION

Absolute zero

Absolute zero (0 K, which is −273 °C) is the lowest possible temperature. It is the temperature at which the molecules of a substance have their **minimum possible kinetic energy**, so the substance has its minimum internal energy.

$$T/\mathrm{K} = \theta/^{\circ}\mathrm{C} + 273$$

Because the kelvin scale starts at absolute zero, the **average kinetic energy of the molecules is directly proportional to the absolute temperature** T. Twice the kelvin temperature means twice the average kinetic energy. Celsius doesn't work like that: going from 20 °C to 40 °C is going from 293 K to 313 K, which increases the average kinetic energy by only about 7%.

COMMON MISTAKE

Always convert to kelvin before using any gas equation. In Celsius, warming a gas from 10 °C to 20 °C looks as if it should double the pressure. In kelvin it is 283 K → 293 K, a rise of only 3.5%.

The ideal gas equation: pV = NkT

Experiments on a fixed amount of gas give three simple rules:

- **Boyle's law:** at constant temperature, $p \propto 1/V$. Squash the gas into half the space and the molecules hit each bit of wall twice as often.
- **Pressure law:** at constant volume, $p \propto T$. Hotter molecules hit the walls harder and more often.
- **Charles's law:** at constant pressure, $V \propto T$.

Pressure is also proportional to the number of molecules N: double the gas in the same box, double the collisions. Put all of these together and you get one equation, where the constant k is the **Boltzmann constant**, $1.38 \times 10^{-23}\ \text{J K}^{-1}$.

Ideal gas equation

FORMULA SHEET

$$pV = NkT$$

p: pressure / Pa V: volume / m³ N: number of molecules k: Boltzmann constant / J K⁻¹
T: absolute temperature / K

An **ideal gas** is one that obeys this equation exactly. Real gases come very close as long as the pressure isn't too high and the temperature is well above the point where the gas would turn into a liquid. The model behind it (the kinetic theory, below) assumes:

- a very large number of molecules moving in random directions with a range of speeds;
- the molecules themselves take up a negligible fraction of the container's volume;
- there are no forces between molecules except during collisions;
- all collisions are elastic, and each lasts a negligible time compared with the time between collisions.

EXAM TIP

For a fixed amount of gas N and k are constant, so $\frac{pV}{T}$ is constant: $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$. Any units of p and V work in this ratio form, as long as they're the same on both sides, but T must be in kelvin.

WORKED EXAMPLE

A helium cylinder of volume 0.050 m³ holds gas at 1.5 × 10⁷ Pa and 20 °C. (a) How many helium atoms does it contain? (b) What is the mass of the helium? (mass of a helium atom = 4.0 u) (c) How many balloons of volume 0.010 m³ at 1.0 × 10⁵ Pa can be filled, if the temperature stays at 20 °C?

Show solution

(a) $T = 20 + 273 = 293\ \text{K}$, so

$$N = \frac{pV}{kT} = \frac{1.5 \times 10^7 \times 0.050}{1.38 \times 10^{-23} \times 293} = 1.85 \times 10^{26}$$

(b) $m = 1.85 \times 10^{26} \times 4.0 \times 1.66 \times 10^{-27} = 1.23\ \text{kg}$

(c) At constant temperature $p_1V_1 = p_2V_2$, so at balloon pressure the gas would take up

$$V_2 = \frac{1.5 \times 10^7 \times 0.050}{1.0 \times 10^5} = 7.5 \text{ m}^3$$

But 0.050 m³ of that has to stay behind in the cylinder (it can't go below balloon pressure), leaving 7.45 m³ for balloons: $7.45 / 0.010 = 745$ balloons.

Core Practical 14: pressure and volume

CORE PRACTICAL

Investigate the relationship between pressure and volume of a gas at fixed temperature

A common set-up is the Boyle's law apparatus: air is trapped in a sealed glass tube above a column of oil. A foot pump pushes more oil in, and a Bourdon gauge reads the pressure. The tube has a constant cross-section, so the length of the air column is proportional to its volume. (A syringe connected to a pressure sensor works the same way.)

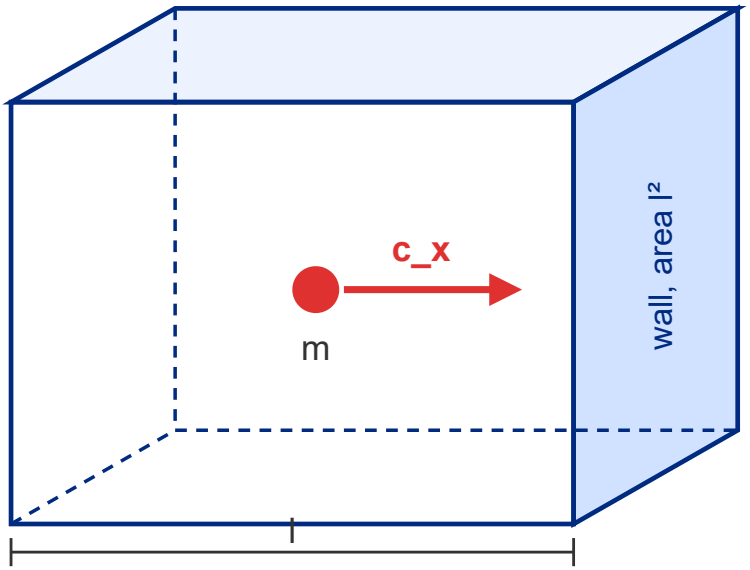
Change the pressure in steps. After each change, **wait** for a minute: compressing a gas warms it, and the law only holds at constant temperature. Then record p and V.

A graph of p against V is a curve, which is hard to test. Plot p **against** $1/V$ instead. From $pV = NkT$, $p = NkT \times \frac{1}{V}$, so the graph should be a **straight line through the origin with gradient** NkT.

- Read the oil level at eye level to avoid parallax.
- Check for leaks: the readings should return to their starting values when you release the pressure.
- Gas in the tubing and the gauge that isn't counted in your volume reading (“dead volume”) is a systematic error. It makes every true volume bigger than the reading.
- **Safety:** the glass tube is under pressure, so wear eye protection or use a safety screen, and never exceed the gauge's maximum pressure.

Kinetic theory: where pressure comes from

Why does a gas push on its container at all? Molecules keep hitting the walls and bouncing off. Every bounce reverses a molecule's momentum, and by Newton's second law a change of momentum needs a force. Countless collisions every second add up to a steady force on each part of the wall, and force per unit area is pressure. Let's turn that picture into an equation.



before: +mc_x

after: −mc_x

Δp = −2mc_x

back again after
Δt = 2l / c_x

One molecule in a cube of side l. Only the velocity component c_x towards the shaded wall matters for collisions with that wall.

Deriving pV = ⅓Nm⟨c²⟩ from molecules hitting the walls

1 A molecule of mass m moves towards the right-hand wall with velocity component c_x. The collision is elastic, so it bounces back with $-c_x$. Its change of momentum is

$$\Delta p = (-mc_x) - (mc_x) = -2mc_x$$

By Newton's third law the wall receives an equal and opposite momentum, $+2mc_x$.

2 Before it hits that wall again it has to cross to the far wall and come back, a distance $2l$, at speed c_x:

$$\Delta t = \frac{2l}{c_x}$$

Bounces off the other walls don't change c_x, so they don't affect this time.

3 The average force this one molecule exerts on the wall is the rate at which it delivers momentum:

$$F = \frac{\Delta p}{\Delta t} = \frac{2mc_x}{2l/c_x} = \frac{mc_x^2}{l}$$

4 There are N molecules, each with its own c_x. Add up their forces and write the sum using the mean value of c_x^2:

$$F = \frac{m}{l}\left(c_{x1}^2 + c_{x2}^2 + \cdots + c_{xN}^2\right) = \frac{Nm\langle c_x^2\rangle}{l}$$

5 Pressure is force divided by area, and the wall's area is l^2. Also $l^3 = V$, the volume of the box:

$$p = \frac{F}{l^2} = \frac{Nm\langle c_x^2\rangle}{l^3} = \frac{Nm\langle c_x^2\rangle}{V}$$

6 Each molecule's speed c satisfies $c^2 = c_x^2 + c_y^2 + c_z^2$ (Pythagoras in 3D). The motion is random, so no direction is special and on average the three components are equal:

$$\langle c_x^2\rangle = \langle c_y^2\rangle = \langle c_z^2\rangle = \tfrac{1}{3}\langle c^2\rangle$$

7 Substitute and multiply both sides by V:

$$p = \frac{Nm\langle c^2\rangle}{3V} \quad \Rightarrow \quad \boxed{pV = \tfrac{1}{3}Nm\langle c^2\rangle}$$

Collisions between molecules don't spoil this: they only swap momentum between molecules, and the random spread of velocities stays the same on average. The cube shape doesn't matter either.

$pV = \frac{1}{3}Nm\langle c^2\rangle$ is **not on the formula sheet**, but you need it: it is the starting point of the derivation that the spec does require, which comes next.

KEY IDEA

What ⟨c²⟩ means

$\langle c^2\rangle$ is the **mean square speed**: square every molecule's speed, then take the average. It is not the same as the mean speed squared. For three molecules moving at 1, 2 and 3 m s⁻¹, the mean speed is 2 m s⁻¹, but $\langle c^2\rangle = (1+4+9)/3 = 4.67\ \text{m}^2\,\text{s}^{-2}$. Its square root is the **root mean square (r.m.s.) speed**, $c_{\text{rms}} = \sqrt{\langle c^2\rangle} = 2.16\ \text{m s}^{-1}$.

Linking molecular speed to temperature

Now we have two equations for pV: one from experiments ($pV = NkT$) and one from a model of bouncing molecules ($pV = \frac{1}{3}Nm\langle c^2\rangle$). Both describe the same gas, so they must agree.

Deriving ½m⟨c²⟩ = 3/2 kT

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 Write down the two expressions for pV: kinetic theory, and the ideal gas equation.

$$pV = \tfrac{1}{3}Nm\langle c^2\rangle \qquad \text{and} \qquad pV = NkT$$

2 They describe the same gas, so set them equal:

$$\tfrac{1}{3}Nm\langle c^2\rangle = NkT$$

3 N appears on both sides, so cancel it, then multiply both sides by 3:

$$m\langle c^2\rangle = 3kT$$

4 Divide both sides by 2, so the left-hand side becomes the average kinetic energy of one molecule:

$$\boxed{\tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT}$$

$\frac{1}{2}m\langle c^2\rangle$ is the mean of $\frac{1}{2}mc^2$ over all molecules: the average (translational) kinetic energy per molecule.

Average kinetic energy of a gas molecule

FORMULA SHEET

$$\tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT$$

m: mass of one molecule / kg $\langle c^2\rangle$: mean square speed / m² s⁻² k: Boltzmann constant / J K⁻¹
T: absolute temperature / K

This is the result behind the earlier statements. The average kinetic energy of a molecule depends **only on the absolute temperature**: not on the pressure, not on the volume, and not on which gas it is. At the same temperature a heavy xenon atom and a light helium atom have the same average kinetic energy, so the helium atom must be moving much faster. At $T = 0$ the average kinetic energy would be zero, which is the idea of absolute zero again.

The r.m.s. speed of the molecules

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 Start from $m\langle c^2\rangle = 3kT$ and divide by m:

$$\langle c^2\rangle = \frac{3kT}{m}$$

2 Take the square root to get the r.m.s. speed:

$$\boxed{c_{\text{rms}} = \sqrt{\langle c^2\rangle} = \sqrt{\frac{3kT}{m}}}$$

So $c_{\text{rms}} \propto \sqrt{T}$: to double the speed of the molecules you need four times the kelvin temperature.

WORKED EXAMPLE

Air is mostly nitrogen molecules, each of mass 28 u. For air at 20 °C, calculate (a) the average kinetic energy of a nitrogen molecule, (b) the r.m.s. speed of the molecules, and (c) the temperature at which the r.m.s. speed would be twice as big.

Show solution

(a) $T = 293$ K:

$$\tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT = 1.5 \times 1.38 \times 10^{-23} \times 293 = 6.07 \times 10^{-21}\ \text{J}$$

(b) $m = 28 \times 1.66 \times 10^{-27} = 4.65 \times 10^{-26}$ kg, so

$$c_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 293}{4.65 \times 10^{-26}}} = 511\ \text{m s}^{-1}$$

That is faster than the speed of sound in air (about 340 m s⁻¹), which makes sense: sound is carried by these molecules.

(c) $c_{\text{rms}} \propto \sqrt{T}$, so doubling the speed needs $4T = 4 \times 293 = 1172$ K, which is about 900 °C.

Quick checks

QUICK CHECK

Ice at 0 °C melts into water at 0 °C. What happens to the molecules?

- Their average kinetic energy increases and their potential energy stays the same.
- Their potential energy increases and their average kinetic energy stays the same.
- Both their kinetic and potential energies stay the same, because the temperature doesn't change.
- Their internal energy decreases as the bonds break.

The temperature is constant, so the average kinetic energy is constant. Energy is still being supplied, so the internal energy rises: it goes into potential energy as the molecules are pulled apart against the forces between them.

QUICK CHECK

A heater of constant power warms some ice and then the water it melts into. Why is the θ–t graph steeper for the ice than for the water?

- Ice has a lower density than water.
- Some of the energy is used to melt the ice.
- Ice has a smaller specific heat capacity than water.
- Water loses more heat to the surroundings.

The gradient is $\Delta\theta/\Delta t = P/mc$. The mass and power are the same for both, and $c_{\text{ice}} \approx 2100$ is about half of $c_{\text{water}} = 4180\ \text{J kg}^{-1}\,\text{K}^{-1}$, so the ice warms about twice as fast.

QUICK CHECK

A gas is heated from 27 °C to 327 °C. By what factor does the r.m.s. speed of its molecules increase?

- 12.1
- 2.0
- 1.41
- 1.10

Convert to kelvin: 300 K → 600 K, so T doubles. $c_{\text{rms}} = \sqrt{3kT/m} \propto \sqrt{T}$, so the speed goes up by $\sqrt{2} = 1.41$. Using Celsius (327/27 = 12.1) is the classic trap.

QUICK CHECK

Helium and xenon are mixed in a container and left to reach the same temperature. Which quantity is the same for both kinds of atom?

- Their r.m.s. speed
- Their momentum
- Their average kinetic energy
- The number of times per second they hit the walls

$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ depends only on T. Xenon atoms are about 33 times more massive, so they move $\sqrt{33} \approx 5.7$ times more slowly, and they hit the walls less often.

QUICK CHECK

A fixed mass of ideal gas has its volume doubled while its kelvin temperature is tripled. What happens to its pressure?

- It falls to 2/3 of its original value.
- It increases to 1.5 times its original value.
- It increases to 6 times its original value.
- It stays the same.

$p = NkT/V$, so $p \propto T/V$. Tripling T and doubling V gives a factor of $3/2 = 1.5$.

QUICK CHECK

An NTC thermistor and a fixed resistor form a potential divider. The output is the p.d. across the fixed resistor. As the temperature rises, what happens to the output?

- It falls, because the thermistor's resistance rises.
- It falls, because the thermistor's resistance falls.
- It rises, because the thermistor's resistance falls.
- It stays constant, because the supply p.d. is constant.

For an NTC thermistor, resistance falls as temperature rises. The fixed resistor then has a bigger share of the total resistance, so it takes a bigger share of the supply p.d.: $V_{\text{out}} = V_{\text{in}}R/(R + R_{\text{th}})$ rises.

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Nuclear Decay

The nucleus holds enormous amounts of energy, and this page is about how it gets out. You will see why a nucleus weighs less than the parts it is made from, why both splitting heavy nuclei and joining light ones release energy, what alpha, beta and gamma radiation are, and why a lump of radioactive material decays in such a predictable way when each nucleus inside it behaves completely at random.

Mass deficit and binding energy

A nucleus is made of protons and neutrons (together called **nucleons**) held together by the strong nuclear force. Weigh a helium-4 nucleus and compare it with two protons plus two neutrons weighed separately, and something odd turns up: the nucleus is **lighter** than its parts. This missing mass is the **mass deficit** (also called the mass defect).

Where has it gone? When nucleons are pulled together by the strong force they lose potential energy, just as a ball loses potential energy when it falls, and that energy is given out. Einstein showed that energy and mass are equivalent, $\Delta E = c^2 \Delta m$, so a system that has given out energy has less mass. To pull the nucleus apart again you would have to put all of that energy back. That energy is the **binding energy**.

DEFINITION

Mass deficit and binding energy

The **mass deficit** of a nucleus is the difference between the total mass of its separate protons and neutrons and the mass of the nucleus itself.

The **binding energy** of a nucleus is the energy needed to separate it completely into its individual protons and neutrons. It equals the energy released when the nucleus forms from them, and $\text{binding energy} = c^2 \times \text{mass deficit}$.

Mass–energy equivalence

FORMULA SHEET

$$\Delta E = c^2 \, \Delta m$$

ΔE: energy / J c: speed of light, 3.00 × 10⁸ m s⁻¹ Δm: change in mass / kg

Because $c^2 = 9.00 \times 10^{16}\ \text{m}^2\,\text{s}^{-2}$ is so huge, a tiny mass is worth a vast amount of energy. One gram of mass is equivalent to 9×10^{13} J, roughly the energy released by the atomic bomb dropped on Hiroshima.

COMMON MISTAKE

Binding energy is **not** energy that the nucleus contains and can give out. It is the energy you would have to **supply** to break the nucleus up. A nucleus with a large binding energy has a large mass deficit: it is in a deep energy “hole”, which makes it more stable, not more energetic.

The atomic mass unit

Nuclear masses are tiny in kilograms, so we use the **atomic mass unit**, u, defined as one twelfth of the mass of a carbon-12 atom:

$$1\ \text{u} = 1.66 \times 10^{-27}\ \text{kg}$$

A proton is 1.00728 u and a neutron 1.00867 u. To convert a mass in u to kg, multiply by 1.66×10^{-27}. The energy equivalent of 1 u, using the data-sheet values, is $c^2 \times 1.66 \times 10^{-27} = 1.49 \times 10^{-10}$ J, which is about 930 MeV (1 MeV = 1.60×10^{-13} J).

EXAM TIP

For a binding-energy calculation: (1) add up the masses of the separate nucleons, (2) subtract the mass of the nucleus to get Δm in u, (3) convert Δm to kg, (4) use $\Delta E = c^2 \Delta m$ to get joules, (5) divide by 1.60×10^{-13} if you want MeV. Keep every decimal place in step 2: you are subtracting two nearly equal numbers, so rounding early wipes out the answer.

WORKED EXAMPLE

A helium-4 nucleus has a mass of 4.00151 u. Calculate (a) its mass deficit in kg, (b) its binding energy in MeV, and (c) its binding energy per nucleon. (proton mass = 1.00728 u, neutron mass = 1.00867 u)

Show solution

Total mass of 2 protons and 2 neutrons:

$$2(1.00728) + 2(1.00867) = 4.03190\ \text{u}$$

(a) $\Delta m = 4.03190 - 4.00151 = 0.03039\ \text{u} = 0.03039 \times 1.66 \times 10^{-27} = 5.04 \times 10^{-29}\ \text{kg}$

(b) $\Delta E = c^2 \Delta m = (3.00 \times 10^8)^2 \times 5.045 \times 10^{-29} = 4.54 \times 10^{-12}\ \text{J}$

$$\frac{4.54 \times 10^{-12}}{1.60 \times 10^{-13}} = 28.4\ \text{MeV}$$

(c) There are 4 nucleons, so the binding energy per nucleon is $28.4/4 = 7.1\ \text{MeV}$.

Binding energy per nucleon: fission and fusion

Big nuclei have more nucleons, so they always have a bigger total binding energy. To compare how tightly different nuclei are held together, divide by the nucleon number. The **binding energy per nucleon** is the average energy needed to remove one nucleon. The higher it is, the more **stable** the nucleus.

Plot it against nucleon number and you get one of the most important graphs in physics. It rises steeply for light nuclei, because each extra nucleon has more neighbours for the strong force to bind it to. It peaks at about 8.8 MeV per nucleon around **iron-56**, the most stable region. Then it falls slowly for heavy nuclei: the strong force only reaches nearest neighbours, but every proton repels every other proton across the whole nucleus, and that repulsion grows as the nucleus gets bigger.

KEY IDEA

Up the curve means energy out

Any nuclear reaction that ends with the nucleons **higher up** the binding-energy-per-nucleon curve releases energy, because the products are more tightly bound and so have less mass.

- **Fusion:** two light nuclei join to make a heavier one, moving up the steep left-hand side.
- **Fission:** a heavy nucleus (such as uranium-235, after absorbing a neutron) splits into two medium-sized nuclei, moving up the gentle right-hand side.

Energy released = total binding energy after − total binding energy before = $c^2 \times$ (mass lost). Neither process can release energy beyond iron: fusing iron or splitting it would both move *down* the curve.

Fusion releases more energy **per nucleon**, because the left-hand side of the curve is so steep. Fission releases more energy **per reaction** (about 200 MeV against 17.6 MeV for deuterium–tritium fusion), because so many nucleons each gain a little.

WORKED EXAMPLE

In a nuclear reactor, uranium-235 absorbs a neutron and splits: ${}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n}$. The binding energies per nucleon are U-235: 7.59 MeV, Ba-141: 8.33 MeV, Kr-92: 8.51 MeV. (a) Calculate the energy released in one fission. (b) Estimate the energy released if 1.0 kg of uranium-235 undergoes fission.

Show solution

(a) Free neutrons have no binding energy, so only the nuclei count.

$$\text{before: } 235 \times 7.59 = 1784 \text{ MeV}$$

$$\text{after: } 141 \times 8.33 + 92 \times 8.51 = 1174.5 + 782.9 = 1957 \text{ MeV}$$

$$\text{energy released} = 1957 - 1784 = 174 \text{ MeV} = 174 \times 1.60 \times 10^{-13} = 2.78 \times 10^{-11} \text{ J}$$

(More energy, bringing the total to about 200 MeV, is released later as the neutron-rich fission products decay.)

(b) Mass of one U-235 nucleus $\approx 235 \times 1.66 \times 10^{-27} = 3.90 \times 10^{-25}$ kg, so 1.0 kg contains

$$N = \frac{1.0}{3.90 \times 10^{-25}} = 2.56 \times 10^{24} \text{ nuclei}$$

$$E = 2.56 \times 10^{24} \times 2.78 \times 10^{-11} = 7.1 \times 10^{13} \text{ J}$$

Burning 1 kg of coal releases about 3×10^{7} J, so this is over two million times more.

Making fusion happen

Fusion sounds easy: push two light nuclei together. The problem is that both are positively charged, so they repel. The strong force that could bind them only works over about 10^{-15} m (1 femtometre, fm), and at that range the electrostatic repulsion is

enormous. Two protons 3 fm apart push on each other with a force of about 26 N, which is huge for something with a mass of 10^{-27} kg.

To get that close, the nuclei must collide with enough kinetic energy to climb this electrostatic “hill”, often called the **Coulomb barrier**. The potential energy at 3 fm is

$$E_{\mathrm{p}} = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r} = \frac{8.99 \times 10^{9} \times (1.60 \times 10^{-19})^2}{3 \times 10^{-15}} = 7.7 \times 10^{-14}\ \mathrm{J} \approx 0.5\ \mathrm{MeV}$$

That is why fusion needs two extreme conditions:

- **A very high temperature** (tens of millions of kelvin or more), so the nuclei move fast enough. The average kinetic energy of a particle is $\frac{3}{2}kT$ (see the Thermodynamics page), so hotter really does mean faster. At these temperatures the electrons are stripped off and the matter is a **plasma** of bare nuclei and electrons.
- **A very high density**, so collisions happen extremely often. Even when it is very hot, only a tiny fraction of collisions are energetic enough to fuse, so you need an enormous number of collisions every second to get a useful rate. The energy released then heats the surrounding nuclei, which keeps the temperature up so that fusion can **continue**.

In a star, gravity squeezes the core to huge densities and holds the hot plasma together. On Earth, no material container can touch plasma at 10^{8} K, so reactors such as tokamaks hold it with strong magnetic fields, and keeping it dense and hot for long enough is the main engineering challenge.

WORKED EXAMPLE

Estimate the temperature needed for two deuterium nuclei to get within 3.0 fm of each other, assuming the energy available in a collision is about $\frac{3}{2}kT$.

Show solution

The energy needed is the electrostatic potential energy at 3.0 fm (each deuterium nucleus has charge $+e$):

$$E_{\mathrm{p}} = \frac{8.99 \times 10^{9} \times (1.60 \times 10^{-19})^2}{3.0 \times 10^{-15}} = 7.67 \times 10^{-14}\ \mathrm{J}$$

Set $\frac{3}{2}kT = E_{\mathrm{p}}$:

$$T = \frac{2E_{\mathrm{p}}}{3k} = \frac{2 \times 7.67 \times 10^{-14}}{3 \times 1.38 \times 10^{-23}} = 3.7 \times 10^{9}\ \mathrm{K}$$

This is only an estimate, but it shows that temperatures of billions of kelvin are needed if every collision has just the average energy.

Alpha, beta and gamma radiation

An unstable nucleus becomes more stable by emitting radiation. There are three main kinds, and the differences between them all follow from what each one is.

Radiation	What it is	Charge	Ionising ability	Range in air	Stopped by
alpha, α	helium nucleus: 2 protons + 2 neutrons, ${}^{4}_{2}\alpha$	+2e	very strong	a few cm	a sheet of paper, or the outer layer of skin
beta-minus, β^-	fast electron emitted from the nucleus, ${}^{0}_{-1}\beta$	−e	moderate	up to about a metre	a few mm of aluminium
gamma, γ	high-frequency electromagnetic radiation (photons)	0	weak	very long; intensity falls with distance	never completely; several cm of lead or metres of concrete reduce it a lot

Why are the strongest ionisers the least penetrating? Each time radiation knocks an electron off an atom it loses a little energy (about 30 eV in air). An alpha particle is heavy, slow and doubly charged, so it interacts with almost every atom it passes, making tens of thousands of ion pairs per centimetre. It spends its few MeV within a few centimetres. A beta particle has less charge and moves much faster, so it interacts less often and gets further. A gamma photon has no charge and only interacts occasionally, so it penetrates a long way but ionises weakly.

That is also why alpha sources are fairly safe outside the body (dead skin stops them) but very dangerous if breathed in or swallowed, where all that ionisation happens in living cells.

Writing nuclear equations

Each nucleus is written ${}^{A}_{Z}\mathrm{X}$: the **nucleon number** A on top and the **proton number** Z underneath. In every nuclear equation, the totals of A and of Z (charge) must be the **same on both sides**.

- **Alpha decay**: the nucleus loses 2 protons and 2 neutrons, so A falls by 4 and Z by 2:

$$ {}^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\alpha $$

- **Beta-minus decay**: a neutron turns into a proton, an electron and an antineutrino ($\mathrm{udd} \rightarrow \mathrm{uud}$ in quark terms). A stays the same and Z goes up by 1:

$$ {}^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\beta + \bar{\nu}_e $$

- **Beta-plus decay**: a proton turns into a neutron, a positron and a neutrino. Z goes down by 1:

$$ {}^{22}_{11}\mathrm{Na} \rightarrow {}^{22}_{10}\mathrm{Ne} + {}^{0}_{+1}\beta + \nu_e $$

- **Gamma emission**: a nucleus left in an excited state (shown with a *) after alpha or beta decay gets rid of its extra energy as a photon. A and Z don't change:

$$ {}^{60}_{28}\mathrm{Ni}^{*} \rightarrow {}^{60}_{28}\mathrm{Ni} + \gamma $$

The same rules apply to fission and fusion. For example, deuterium–tritium fusion is ${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$: the top numbers give 5 = 5 and the bottom ones 2 = 2.

WORKED EXAMPLE

Uranium-238 ($Z = 92$) decays by emitting an alpha particle, and the product then emits two beta-minus particles. (a) Identify the final nucleus. (b) Uranium-238 eventually becomes stable lead-206 ($Z = 82$). How many alpha and beta-minus decays happen along the way?

Show solution

(a) Alpha: $A = 238 - 4 = 234$, $Z = 92 - 2 = 90$ (thorium-234). Each beta-minus adds 1 to Z: $90 \to 91 \to 92$. The final nucleus is $^{234}_{92}\mathrm{U}$, a different isotope of uranium.

(b) Only alpha decay changes A, by 4 each time: $238 - 206 = 32$, so there are $32/4 = 8$ alpha decays. Those would reduce Z by 16, from 92 to 76. To end at 82, Z must rise by 6, so there are **6 beta-minus decays**.

Background radiation

Switch on a Geiger–Müller (GM) tube with no source anywhere near it and it still clicks every second or two. That is **background radiation**, and it comes from:

- **radon gas** seeping out of rocks and soil (in many places the largest part);
- rocks, soil and building materials such as granite and concrete;
- **cosmic rays** from space, and the particles they create in the atmosphere;
- food and drink, which contain small amounts of radioactive potassium-40 and carbon-14, so your own body is slightly radioactive;
- medical uses such as X-rays and scans, and a small amount from nuclear power and past weapons tests.

Background is random and changes from place to place, so you can't just guess it. Before an experiment, remove all sources and **measure the background count over a long time** (several minutes), then work out the background count rate. Subtract it from every reading to get the **corrected count rate**, the part that comes from your source.

$$\text{corrected count rate} = \text{measured count rate} - \text{background count rate}$$

COMMON MISTAKE

Subtract count **rates**, not raw counts taken over different times. Convert both to counts per second (or both to counts per minute) first. Background matters most when the source count rate is low, for example behind thick lead or after many half-lives.

WORKED EXAMPLE

With no source present, a GM tube records 186 counts in 10 minutes. A source is then placed in front of it and the tube records 1240 counts in 2 minutes. Calculate the corrected count rate in counts per second.

Show solution

Background count rate $= 186/600 \text{ s} = 0.31 \text{ s}^{-1}$.

Measured count rate $= 1240/120 \text{ s} = 10.33 \text{ s}^{-1}$.

Corrected count rate $= 10.33 - 0.31 = 10.0 \text{ s}^{-1}$.

Core Practical 15: absorption of gamma radiation

CORE PRACTICAL

Investigate the absorption of gamma radiation by lead

First measure the **background count rate**: with the source locked away, count for about 10 minutes with the GM tube and counter. Then clamp a gamma source (such as cobalt-60) and the GM tube a fixed distance apart, so nothing moves during the experiment. Count for a fixed time (100 s or more) with no lead in between. Then put lead sheets of increasing thickness x between them, measuring each thickness with a micrometer at several points and averaging, and count again each time.

Subtract the background from every reading and plot **corrected count rate against thickness of lead**. The count rate falls quickly at first and then more and more slowly. It never reaches zero: lead reduces gamma radiation but never stops all of it.

- Decay is random, so counts fluctuate. Count for long enough to get several hundred counts (the uncertainty in a count N is about $\sqrt{N}$), and repeat and average.
- Keep the distance fixed: gamma intensity also falls with distance from the source.
- **Safety:** handle the source with long tongs, never with your hands. Point it away from people, keep it at arm's length, keep the time it is out of its lead-lined box short, and put it straight back afterwards. Lead is toxic, so wear gloves or wash your hands after handling the sheets.

Random decay and half-life

Radioactive decay is **spontaneous**: nothing outside the nucleus triggers it, and it isn't affected by temperature, pressure or chemical reactions. It is also **random**: there is no way to predict which nucleus will decay next, or when. Every undecayed nucleus of a given isotope has the **same chance** of decaying in the next second, no matter how long it has already survived. A nucleus doesn't "get old".

Think of each nucleus as a die, rolled once every second, that decays if it shows a six. You can't say when any one die will roll a six. But roll 6000 dice and you can be confident about 1000 sixes will come up. In the same way, if each nucleus has a probability λ of decaying per second, then out of N nuclei about λN decay each second. With the huge numbers of nuclei in any real sample, this prediction is extremely accurate.

DEFINITION

Activity, decay constant and half-life

The **activity** A of a sample is the number of nuclei that decay per unit time. Its unit is the becquerel: 1 Bq = 1 decay per second.

The **decay constant** λ is the probability that a nucleus decays per unit time. Its unit is s^{-1}.

The **half-life** $t_{1/2}$ of an isotope is the time taken for half of the undecayed nuclei in a sample to decay (equivalently, for the activity to halve).

Activity

FORMULA SHEET

$$A = \lambda N$$

A: activity / Bq λ: decay constant / s⁻¹
N: number of undecayed nuclei

Rate of decay

FORMULA SHEET

$$\frac{dN}{dt} = -\lambda N$$

dN/dt: rate of change of N / s⁻¹

$\frac{dN}{dt}$ means the rate at which N changes. It is negative because N goes down, and it is proportional to N: the more nuclei there are, the faster they disappear. As N falls, the rate of decay falls with it. The result is **exponential decay**: in every equal interval of time, the same **fraction** of the remaining nuclei decays. This is the same maths as a capacitor discharging through a resistor.

EXAM TIP

To find a half-life from an N–t or A–t graph, read the starting value, find where it has halved and read the time. Then check from there to a quarter, and from a quarter to an eighth: each should take the same time. Averaging several of these gives a more reliable value. If you are given count rates, make sure background has been subtracted first.

The exponential decay equations

The solution of $dN/dt = -\lambda N$ is an exponential. Since $A = \lambda N$, the activity decays in exactly the same way:

Number of undecayed nuc

FORMULA SHEET

$$N = N_0\, e^{-\lambda t}$$

N_0: number at $t = 0$

Activity

FORMULA SHEET

$$A = A_0\, e^{-\lambda t}$$

A_0: activity at $t = 0$ / Bq

Decay constant and half-lif FORMULA SHEET

$$\lambda = \frac{\ln 2}{t_{1/2}}$$

$t_{1/2}$: half-life / s

Where λ = ln 2 / t½ comes from DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. After one half-life, half of the nuclei are left: $N = \frac{1}{2}N_0$ when $t = t_{1/2}$. Put this into $N = N_0 e^{-\lambda t}$:

$$\tfrac{1}{2}N_0 = N_0\, e^{-\lambda t_{1/2}} \quad \Rightarrow \quad \tfrac{1}{2} = e^{-\lambda t_{1/2}}$$

2. Take the natural log of both sides. $\ln \frac{1}{2} = -\ln 2$, and $\ln(e^x) = x$:

$$-\ln 2 = -\lambda t_{1/2} \quad \Rightarrow \quad \boxed{\lambda = \frac{\ln 2}{t_{1/2}}}$$

So a large decay constant means a short half-life. $\ln 2 \approx 0.693$.

An exponential curve is hard to read accurately, so we turn it into a straight line by taking logs. The spec says you must be able to derive these log equations.

Deriving the log equations: ln N = ln N₀ − λt SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Start from the decay equation:

$$N = N_0\, e^{-\lambda t}$$

2. Take the natural log of both sides:

$$\ln N = \ln\left(N_0\, e^{-\lambda t}\right)$$

3. The log of a product is the sum of the logs, $\ln(ab) = \ln a + \ln b$:

$$\ln N = \ln N_0 + \ln\left(e^{-\lambda t}\right)$$

4. $\ln$ undoes e, so $\ln(e^{-\lambda t}) = -\lambda t$:

$$\boxed{\ln N = \ln N_0 - \lambda t}$$

5 The same steps applied to $A = A_0 e^{-\lambda t}$ give the log equation for activity (and for count rate, once background has been subtracted):

$$\boxed{\ln A = \ln A_0 - \lambda t}$$

Compare with $y = mx + c$: plotting $\ln N$ (or $\ln A$) on the y-axis against t gives a straight line with gradient $-\lambda$ and y-intercept $\ln N_0$.

WORKED EXAMPLE

A hospital has a sample of iodine-131 of mass 2.0 μg. Iodine-131 has a half-life of 8.0 days. (a) Show that the sample contains about 9×10^{15} nuclei. (b) Calculate its activity. (c) Calculate its activity 30 days later.

Show solution

(a) Mass of one nucleus $\approx 131 \times 1.66 \times 10^{-27} = 2.17 \times 10^{-25}$ kg, so

$$N = \frac{2.0 \times 10^{-9}}{2.17 \times 10^{-25}} = 9.2 \times 10^{15}$$

(b) The half-life must be in seconds for an activity in Bq: $t_{1/2} = 8.0 \times 24 \times 3600 = 6.91 \times 10^5$ s.

$$\lambda = \frac{\ln 2}{6.91 \times 10^5} = 1.00 \times 10^{-6}\ \text{s}^{-1}, \qquad A = \lambda N = 1.00 \times 10^{-6} \times 9.2 \times 10^{15} = 9.2 \times 10^9\ \text{Bq}$$

(c) In days, $\lambda = \ln 2 / 8.0 = 0.0866\ \text{day}^{-1}$, so $\lambda t = 0.0866 \times 30 = 2.60$:

$$A = A_0 e^{-\lambda t} = 9.2 \times 10^9 \times e^{-2.60} = 6.8 \times 10^8\ \text{Bq}$$

Notice you can use any time unit for λt, as long as λ and t use the same one.

WORKED EXAMPLE

Carbon-14 has a half-life of 5730 years. A piece of ancient wood has a corrected count rate that is 30% of the rate from the same mass of living wood. Estimate the age of the wood.

Show solution

$\lambda = \ln 2 / 5730 = 1.21 \times 10^{-4}\ \text{year}^{-1}$. Using the log equation with $A/A_0 = 0.30$:

$$\ln A = \ln A_0 - \lambda t \quad \Rightarrow \quad t = \frac{\ln(A_0/A)}{\lambda} = \frac{\ln(1/0.30)}{1.21 \times 10^{-4}} = 9.95 \times 10^3\ \text{years}$$

Check: 30% is between $\frac{1}{2}$ (one half-life) and $\frac{1}{4}$ (two half-lives), and 9950 years (about 10 000 years) is between 5730 and 11 460 years.

Quick checks

QUICK CHECK

A radioactive sample is heated from 20 °C to 200 °C. What happens to its half-life?

- It decreases, because the nuclei have more energy.
- It increases, because the sample expands.
- It stays the same.
- It decreases at first and then returns to normal.

Decay is spontaneous: it isn't affected by temperature, pressure or chemistry. Ordinary heating changes the motion of whole atoms, which is nowhere near enough to affect what happens inside the nucleus.

QUICK CHECK

What fraction of a radioactive isotope remains undecayed after three half-lives?

- 1/3
- 1/6
- 1/8
- 1/9

Each half-life halves what is left: $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$. It never drops to zero after a whole number of half-lives.

QUICK CHECK

Why do alpha particles have such a short range in air?

- They are absorbed by the nitrogen nuclei in the air.
- **They are strongly ionising, so they lose their energy in many interactions over a short distance.**
- They have no charge, so they are not affected by the air.
- They are emitted with much less energy than beta particles.

Alpha particles are heavy, slow and have charge +2e, so they ionise almost every atom they pass. Each ionisation takes some energy, so their few MeV are used up within a few centimetres. (Alpha particles are usually emitted with *more* energy than beta particles, not less.)

QUICK CHECK

Which of these reactions releases energy?

- Splitting an iron-56 nucleus into two smaller nuclei
- **Joining two hydrogen-2 nuclei to make helium**
- Joining two uranium-235 nuclei together
- Splitting a helium-4 nucleus into two hydrogen-2 nuclei

Energy is released when the products have a higher binding energy per nucleon. Fusing hydrogen-2 (about 1.1 MeV per nucleon) into helium-4 (7.1 MeV per nucleon) moves steeply up the curve. The other three all move down the curve, so they would need energy to be supplied.

QUICK CHECK

Why must the fuel in a fusion reactor be at a very high temperature?

- So the nuclei can release their electrons and become neutral.
- So the strong force between nuclei becomes stronger.
- **So the nuclei have enough kinetic energy to overcome their electrostatic repulsion and get close enough to fuse.**
- So the fuel has a higher density.

Nuclei are positive and repel. They have to get within about $10^{-15}\ \text{m}$ for the short-range strong force to act, which needs enormous kinetic energy, and so an enormous temperature. (High density is needed too, but for a different reason: to make collisions frequent enough.)

QUICK CHECK

A graph of ln(corrected count rate) against time is a straight line with gradient −0.0231 s^{-1}. What is the half-life?

- 0.016 s
- **30 s**
- 43 s
- 0.0231 s

The gradient is $-\lambda$, so $\lambda = 0.0231\ \text{s}^{-1}$ and $t_{1/2} = \ln 2/\lambda = 0.693/0.0231 = 30\ \text{s}$. (43 s is $1/\lambda$, which is not the half-life.)

QUICK CHECK

Polonium-210 ($Z = 84$) decays by alpha emission. What is the product?

- $^{210}_{82}\mathrm{Pb}$
- ✓ $^{206}_{82}\mathrm{Pb}$
- $^{206}_{84}\mathrm{Po}$
- $^{210}_{85}\mathrm{At}$

Alpha emission removes $^{4}_{2}\alpha$: $A = 210 - 4 = 206$ and $Z = 84 - 2 = 82$, which is lead. Check: $210 = 206 + 4$ and $84 = 82 + 2$.

A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Oscillations

Pendulums, guitar strings, car suspensions, the atoms in a crystal and the swaying of a skyscraper all oscillate. Most of them follow the same simple rule, simple harmonic motion, and once you understand that one rule you can predict their periods, their speeds and their energies. Then we look at what happens when oscillations are damped, and when something pushes them at just the right frequency.

What makes motion simple harmonic?

Every oscillator has an **equilibrium position** where it would happily sit still. Pull it away and a **restoring force** pushes it back. It arrives back at equilibrium moving fast, overshoots because of its inertia, slows down on the other side, and comes back again.

The motion is **simple harmonic** when the restoring force is **proportional to the displacement**: pull twice as far, get twice the force back. That is exactly what a spring obeying Hooke's law does.

The condition for simple harmonic motion

FORMULA SHEET

$$F = -kx$$

F: restoring force / N k: constant (the stiffness) / N m^{-1} x: displacement from equilibrium / m

The **minus sign** is the important part: the force always points the opposite way to the displacement, back towards equilibrium. Since $F = ma$, the acceleration behaves in the same way: it is proportional to the displacement and always directed towards the equilibrium position.

DEFINITION

Simple harmonic motion

Simple harmonic motion (SHM) is an oscillation in which the acceleration of the object is **directly proportional to its displacement** from the equilibrium position and is **always directed towards** the equilibrium position.

Spotting SHM

To decide whether something moves with SHM, work out the resultant force on it at a displacement x. If it comes out as $F = -(\text{a constant}) \times x$, the motion is simple harmonic.

System	SHM?	Why
mass on a spring	yes	Hooke's law gives $F = -kx$, as long as the spring stays within its limit of proportionality
simple pendulum	yes, for small swings	the restoring force is $-mg\sin\theta$, which is close to proportional to x only when θ is small
a floating block pushed down and released	yes	the extra upthrust is proportional to how far it is pushed down
a ball bouncing on the floor	no	while it is in the air the force on it is mg, the same at every height, not proportional to its displacement

The equations of SHM

Combining $F = -kx$ with $F = ma$ gives $a = -(k/m)x$. The constant k/m is always positive, so we write it as ω^2, and ω turns out to be the **angular frequency** of the oscillation.

Defining equation of S

FORMULA SHEET

$$a = -\omega^2 x$$

a: acceleration / m s^{-2}

ω: angular frequency / rad s^{-1}

Period, frequency and

FORMULA SHEET

$$T = \frac{1}{f} = \frac{2\pi}{\omega}, \qquad \omega = 2\pi f$$

T: period / s f: frequency / Hz

If you let go of the oscillator at $x = A$ (its **amplitude**, the maximum displacement) at $t = 0$, its displacement, velocity and acceleration at any time are:

Displacement

FORMULA SHEET

$$x = A\cos\omega t$$

Velocity

FORMULA SHEET

$$v = -A\omega\sin\omega t$$

Acceleration

FORMULA SHEET

$$a = -A\omega^2 \cos \omega t$$

Compare the first and last: $a = -\omega^2(A \cos \omega t) = -\omega^2 x$, the defining equation again. The biggest speed is $v_{\max} = A\omega$, reached as the object passes through equilibrium. The biggest acceleration is $a_{\max} = A\omega^2$, at the two ends of the motion.

Where the cosine and the ω come from

Picture a point moving steadily round a circle of radius A with angular velocity ω (as in circular motion). After time t it has turned through an angle $\theta = \omega t$. Shine a light so that its shadow falls on a diameter: the shadow moves back and forth with displacement $A \cos \omega t$, which is SHM. The point's speed round the circle is $A\omega$, and its centripetal acceleration $A\omega^2$ points to the centre. Take the components of these along the diameter and you get exactly $v = -A\omega \sin \omega t$ and $a = -A\omega^2 \cos \omega t$. So SHM is “circular motion seen side-on”, and ω is the angular velocity of that imaginary point.

EXAM TIP

Put your calculator in **radians** for $\cos \omega t$ and $\sin \omega t$: ωt is an angle in radians. Also, $x = A \cos \omega t$ only applies if timing starts at maximum displacement. If timing starts as the object passes through equilibrium, the displacement is $x = A \sin \omega t$ instead.

Displacement, velocity and acceleration graphs

The three graphs are linked by gradients, just as in the Motion topic:

- the **gradient of the displacement–time graph is the velocity**. Where x is at a peak or a trough the graph is flat, so $v = 0$. Where x crosses zero the graph is steepest, so the speed is greatest;
- the **gradient of the velocity–time graph is the acceleration**. So a is zero where v is at its biggest, and largest where v is zero.

All three are sine or cosine curves with the same period. The velocity is a quarter of a cycle out of step with the displacement, and the acceleration is exactly half a cycle out of step: it is the displacement graph turned upside down and stretched by ω^2, because $a = -\omega^2 x$.

Position	Displacement	Velocity	Acceleration

at either end	$\pm A$ (maximum)	zero	maximum, $\mp A\omega^2$, pointing to the middle
passing through equilibrium	zero	maximum, $\pm A\omega$	zero

Mass–spring systems and pendulums

For a particular oscillator, ω comes from comparing its force equation with $a = -\omega^2 x$. That gives the two standard results.

Period of a mass on a spring

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A mass m on a spring of stiffness k feels a restoring force $F = -kx$. Newton's second law, $F = ma$, gives

$$ma = -kx \quad \Rightarrow \quad a = -\frac{k}{m}x$$

2. This has the form $a = -\omega^2 x$, so the motion is SHM with

$$\omega^2 = \frac{k}{m} \quad \Rightarrow \quad \omega = \sqrt{\frac{k}{m}}$$

3. Use $T = 2\pi/\omega$:

$$\boxed{T = 2\pi\sqrt{\frac{m}{k}}}$$

A heavier mass has more inertia, so it oscillates more slowly. A stiffer spring pulls back harder, so it oscillates faster. The amplitude doesn't appear at all.

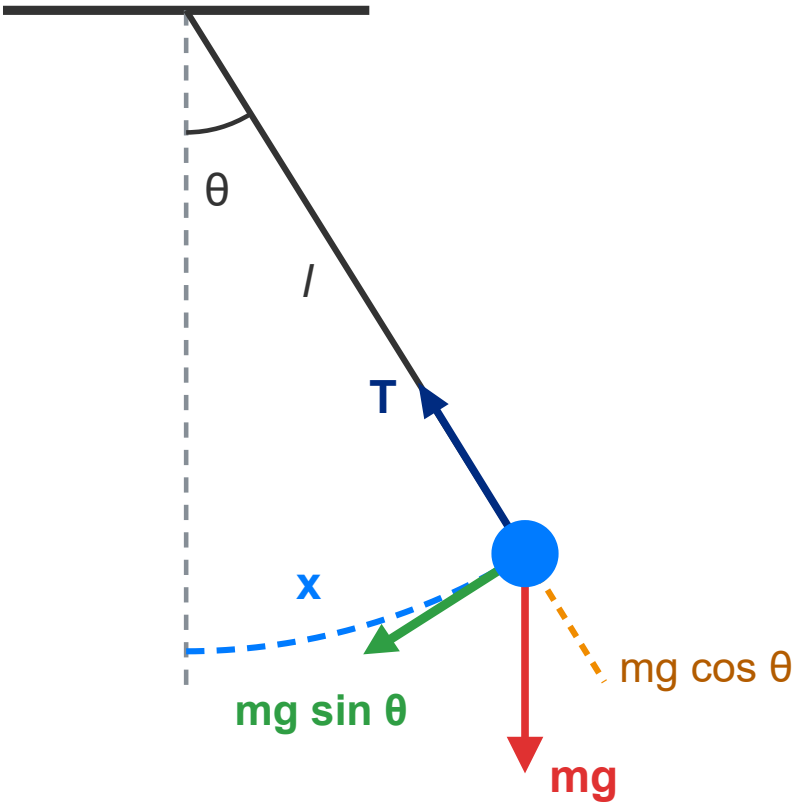
Mass–spring oscillator

FORMULA SHEET

$$T = 2\pi\sqrt{\frac{m}{k}}$$

m: oscillating mass / kg k: spring constant / N m^{-1}

The same formula works for a spring hanging vertically. Gravity just moves the equilibrium position down; the extra force for each extra bit of stretch is still kx.



The component of the weight along the arc, $mg\sin\theta$, pulls the bob back towards equilibrium. The tension and $mg\cos\theta$ act along the string and don't change its speed along the arc.

Period of a simple pendulum (small angles)

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A bob of mass m on a string of length l is pulled aside by an angle θ. The component of its weight along the arc pulls it back towards equilibrium:

$$F = -mg\sin\theta$$

2. For **small angles** (in radians), $\sin\theta \approx \theta$. The bob's displacement along the arc is $x = l\theta$, so $\theta = x/l$:

$$F \approx -mg\theta = -\frac{mg}{l}\,x$$

At 10°, $\sin\theta$ and θ differ by only 0.5%, so this works well for swings of up to about 10°.

3. Divide by m to get the acceleration. The mass cancels:

$$a = -\frac{g}{l}\,x \quad \Rightarrow \quad \omega^2 = \frac{g}{l}$$

4. Use $T = 2\pi/\omega$:

$$\boxed{T = 2\pi\sqrt{\frac{l}{g}}}$$

The period depends only on the length and on g: not on the mass of the bob, and not on the amplitude while the swing stays small.

Simple pendulum

FORMULA SHEET

$$T = 2\pi\sqrt{\frac{l}{g}}$$

l: length from pivot to centre of bob / m g: gravitational field strength / N kg^{-1}

EXAM TIP

When you time a pendulum, time 10 or 20 complete oscillations and divide, so the reaction-time error is shared out. Start and stop the timer as the bob passes the **centre**, where it moves fastest, using a marker behind it.

WORKED EXAMPLE

A 0.40 kg mass hangs on a spring with $k = 60\ \text{N m}^{-1}$. It is pulled down 5.0 cm and released at $t = 0$. Calculate (a) the period, (b) the maximum speed, (c) the maximum acceleration, and (d) the displacement and velocity at $t = 0.10\ \text{s}$.

Show solution

(a) $T = 2\pi\sqrt{m/k} = 2\pi\sqrt{0.40/60} = 0.513\ \text{s}$, and $\omega = \sqrt{k/m} = \sqrt{150} = 12.2\ \text{rad s}^{-1}$.

(b) $v_{\max} = A\omega = 0.050 \times 12.25 = 0.61\ \text{m s}^{-1}$

(c) $a_{\max} = A\omega^2 = 0.050 \times 150 = 7.5\ \text{m s}^{-2}$

(d) $\omega t = 12.25 \times 0.10 = 1.225\ \text{rad}$ (radians!), so

$$x = A\cos\omega t = 0.050\cos(1.225) = 0.017\ \text{m} = 1.7\ \text{cm}$$

$$v = -A\omega\sin\omega t = -0.612 \times \sin(1.225) = -0.58\ \text{m s}^{-1}$$

The minus sign means it is moving back towards (and about to pass through) the equilibrium position, which makes sense: 0.10 s is just under a quarter of a period.

WORKED EXAMPLE

A student times 20 oscillations of a pendulum of length 0.800 m and gets 35.9 s. Calculate a value for g.

Show solution

$T = 35.9/20 = 1.795\ \text{s}$. Squaring $T = 2\pi\sqrt{l/g}$ and rearranging:

$$g = \frac{4\pi^2 l}{T^2} = \frac{4\pi^2 \times 0.800}{1.795^2} = 9.80\ \text{N kg}^{-1}$$

A better method is to use several lengths and plot T^2 against l: the gradient is $4\pi^2/g$.

Energy in SHM

As an oscillator moves, energy keeps changing form. For a mass on a spring it swaps between **kinetic energy** of the mass and **elastic potential energy** stored in the spring (for a pendulum, gravitational potential energy). If

nothing dissipates energy, the total stays constant: the oscillation is **undamped**.

- At the ends ($x = \pm A$) the mass is momentarily at rest: all the energy is potential.
- At equilibrium ($x = 0$) the spring has its natural stretch: all the energy is kinetic, and the speed is $v_{\max}$.

Total energy of a mass–spring oscillator

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. The elastic potential energy stored in a spring is $\frac{1}{2}F\Delta x$. With $F = kx$, at displacement x it stores

$$E_{\mathrm{p}} = \tfrac{1}{2}kx^2$$

2. At the end of the swing, $x = A$ and the mass is at rest, so all the energy is potential:

$$\boxed{E_{\mathrm{total}} = \tfrac{1}{2}kA^2}$$

3. Check at equilibrium, where all of it is kinetic. Using $v_{\max} = A\omega$ and $\omega^2 = k/m$:

$$\tfrac{1}{2}mv_{\max}^2 = \tfrac{1}{2}mA^2\omega^2 = \tfrac{1}{2}mA^2\frac{k}{m} = \tfrac{1}{2}kA^2$$

The two agree, as energy conservation says they must. The energy is proportional to A^2: double the amplitude and the oscillator holds four times as much energy.

At any point in between, $E_{\mathrm{k}} + E_{\mathrm{p}} = \frac{1}{2}kA^2$. Because both depend on squares, each one goes through two maxima per oscillation: kinetic and potential energy oscillate at **twice** the frequency of the motion.

WORKED EXAMPLE

For the 0.40 kg mass on the 60 N m⁻¹ spring with amplitude 5.0 cm, find (a) the total energy and (b) the speed of the mass when its displacement is 3.0 cm.

Show solution

(a) $E = \frac{1}{2}kA^2 = 0.5 \times 60 \times 0.050^2 = 0.075$ J

(b) At $x = 0.030$ m: $E_{\mathrm{p}} = 0.5 \times 60 \times 0.030^2 = 0.027$ J, so $E_{\mathrm{k}} = 0.075 - 0.027 = 0.048$ J.

$$v = \sqrt{\frac{2E_{\mathrm{k}}}{m}} = \sqrt{\frac{2 \times 0.048}{0.40}} = 0.49 \text{ m s}^{-1}$$

Damping

Real oscillations die away. Air resistance, friction and the slight heating of the spring as it flexes all do work against the motion, so energy is transferred out of the oscillator, mostly as thermal energy in the surroundings. Forces like this cause **damping**, and the amplitude gets smaller and smaller.

DEFINITION

Damping

Damping is the effect of **dissipative forces** (such as friction and air resistance) that transfer energy away from an oscillating system, so the amplitude of the oscillations decreases.

- **Light damping:** the amplitude shrinks gradually over many oscillations, and the period stays almost the same (a pendulum in air).
- **Heavy damping:** the system creeps slowly back to equilibrium without oscillating at all (a pendulum in thick oil).
- In between, **critical damping** returns the system to equilibrium in the shortest time without overshooting. Car suspensions are designed to be close to this, so the car doesn't keep bouncing after a bump.

Energy is still conserved in a damped oscillator. It is just shared with the surroundings: kinetic + potential energy of the oscillator + energy dissipated = the energy it started with, $\frac{1}{2}kA_0^2$. Each cycle the oscillator loses a fraction of its energy, which is why the amplitude keeps falling.

Plastic deformation as a damper

A ductile material such as steel, lead or copper can be stretched or bent permanently (plastic deformation). When that happens, the work done to deform it is **not given back** when the force is removed: it ends up as thermal energy inside the material. On a force–extension graph, the energy dissipated in each loading and unloading cycle is the area enclosed between the two curves.

Engineers use this to protect buildings in earthquake zones. Dampers made of ductile metal (and bearings with lead cores) are built into the structure so that they deform plastically as the building sways. Each sway takes energy out of the oscillation, so the amplitude falls quickly instead of building up to a level that would damage the main structure.

Free and forced oscillations: resonance

DEFINITION

Free and forced oscillations

A **free oscillation** happens when a system is displaced and then left alone, with no external periodic force. It oscillates at its own **natural frequency**, f_0.

A **forced oscillation** happens when a periodic driving force acts on the system. Once it has settled, the system oscillates at the **driving frequency**, whatever its natural frequency is.

Resonance occurs when the driving frequency equals the natural frequency of the system. The amplitude of the oscillations is then a maximum, because energy is transferred from the driver to the oscillator most efficiently.

Think of pushing a child on a swing. Push once per swing, at the right moment, and every push is in the same direction as the swing is moving. Each push does positive work, so energy builds up cycle after cycle. The amplitude only stops growing when the energy lost to damping in each cycle equals the energy the pushes add. Push at the wrong frequency and some pushes act against the motion, taking energy back out, so the swing never gets far.

How the amplitude of a forced oscillation depends on the driving frequency:

- **Well below** f_0: the system simply follows the driver, with roughly the driver's amplitude.
- **At** f_0: resonance, and the amplitude is at its peak.
- **Well above** f_0: the system can't keep up and the amplitude falls towards zero.

Damping changes the shape of this resonance curve. More damping makes the peak **lower** and **broader** (the system responds over a wider range of frequencies, but never very strongly), and the peak moves slightly **below** f_0. Very light damping gives a tall, sharp peak.

Resonance in real life

Resonance is useful in musical instruments (the air in a guitar body or a flute resonates), in tuning a radio to one station, and in the quartz crystal that keeps a watch on time. It can also be destructive. A building whose natural frequency matches the shaking of an earthquake can sway with a huge amplitude, which is why tall buildings are fitted with dampers. In 2000 the Millennium Bridge in London began swaying sideways as soon as it opened, because the walkers fell into step with its natural frequency, and it had to be closed until dampers were fitted.

WORKED EXAMPLE

A tall building in an earthquake zone has a natural frequency of 0.8 Hz. Explain why it could be badly damaged by an earthquake with ground vibrations of about 0.8 Hz, and why engineers fit dampers.

Show solution

The ground shaking is a periodic driving force, so the building performs forced oscillations. When the driving frequency (0.8 Hz) equals the building's natural frequency, resonance occurs: energy is transferred to the building most efficiently, so the amplitude of its sway becomes very large and can exceed what the structure can withstand.

Dampers dissipate energy from the oscillations (for example by plastic deformation of ductile metal, or by friction in fluid dampers). This makes the resonance peak much lower, so the maximum amplitude is smaller, and it makes the oscillations die away quickly once the shaking stops.

Core Practical 16: finding an unknown mass

CORE PRACTICAL

Determine the value of an unknown mass using the resonant frequencies of the oscillation of known masses

Hang a spring from a vibration generator connected to a signal generator, and attach a known mass m. Slowly increase the driving frequency and find the frequency at which the mass oscillates with the **largest amplitude**. That is resonance, so the driving frequency equals the natural frequency of the mass–spring system. Repeat for several known masses.

From $T = 2\pi\sqrt{m/k}$, squaring gives

$$T^2 = \frac{4\pi^2}{k}\, m$$

So with $T = 1/f$, a graph of T^2 **against** m is a straight line with gradient $4\pi^2/k$. Now hang the unknown mass, find its resonant frequency, work out T^2, and read its mass from the graph (or calculate $m = T^2/\text{gradient}$).

- Find each resonant frequency by approaching it from below and from above, and take the mean. Fine-tune the frequency in small steps near the peak.
- Keep the driving amplitude small, so the spring stays within its limit of proportionality and never goes slack.
- The spring has mass too, and part of it oscillates. This is a systematic error that gives the line a small intercept, but using the graph (rather than a single calculation) deals with it, because the unknown mass is read from the same line.
- **Safety:** clamp the stand firmly to the bench (or use a counterweight) so it can't topple, and keep your feet clear in case a mass falls.

WORKED EXAMPLE

A student's graph of T^2 against m has a gradient of 1.58 s² kg⁻¹. An unknown mass on the same spring resonates at 1.40 Hz. Calculate (a) the spring constant and (b) the unknown mass.

Show solution

(a) The gradient is $4\pi^2/k$, so

$$k = \frac{4\pi^2}{1.58} = 25.0 \text{ N m}^{-1}$$

(b) $T = 1/f = 1/1.40 = 0.714$ s, so $T^2 = 0.510$ s^2 and

$$m = \frac{T^2}{\text{gradient}} = \frac{0.510}{1.58} = 0.32 \text{ kg}$$

Quick checks

QUICK CHECK

Which equation describes an object moving with simple harmonic motion?

- $a = \omega^2 x$
- $a = -\omega^2 x$
- $a = -\omega x^2$
- $a = -g$

The acceleration must be proportional to the displacement (x, not x^2) and opposite to it, hence the minus sign. $a = -g$ is a constant acceleration, like a bouncing ball in the air, which is not SHM.

QUICK CHECK

An object in SHM passes through its equilibrium position. Which statement is correct?

- Its velocity is zero and its acceleration is a maximum.
- Both its velocity and its acceleration are a maximum.
- Its speed is a maximum and its acceleration is zero.
- Both its velocity and its acceleration are zero.

At $x = 0$, $a = -\omega^2 x = 0$: there is no restoring force there. All of the energy is kinetic, so the speed is at its greatest, $A\omega$.

QUICK CHECK

A pendulum clock is taken to the Moon, where g is one sixth of its value on Earth. By what factor does the period change?

- It becomes 6 times longer.
- It becomes about 2.4 times longer.
- It becomes about 2.4 times shorter.
- It doesn't change, because the period doesn't depend on mass.

$T = 2\pi\sqrt{l/g}$, so $T \propto 1/\sqrt{g}$. Dividing g by 6 multiplies T by $\sqrt{6} = 2.45$. The period doesn't depend on the bob's mass, but it does depend on g.

QUICK CHECK

The amplitude of a mass–spring oscillator is doubled. What happens to its period and to its total energy?

- The period doubles and the energy doubles.
- The period is unchanged and the energy doubles.
- The period is unchanged and the energy is four times bigger.
- The period doubles and the energy is four times bigger.

$T = 2\pi\sqrt{m/k}$ has no A in it. The total energy is $\frac{1}{2}kA^2$, so doubling A multiplies it by 4.

QUICK CHECK

The damping of a driven oscillator is increased. How does its resonance curve (amplitude against driving frequency) change?

- The peak becomes taller and narrower.
- The peak becomes lower and broader.
- The peak moves to a much higher frequency.
- The curve doesn't change, only how quickly it is reached.

More damping dissipates more energy per cycle, so the amplitude at resonance is smaller and the response is spread over a wider range of frequencies. The peak also moves slightly *below* the natural frequency, not above it.

QUICK CHECK

A mass on a spring has a period of 0.50 s. The mass is replaced by one four times as heavy. What is the new period?

- 0.25 s
- 0.50 s
- 1.0 s
- 2.0 s

$T \propto \sqrt{m}$, so four times the mass gives $\sqrt{4} = 2$ times the period: 1.0 s.

A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Gravitational Fields

Every mass in the universe pulls on every other mass. That pull keeps your feet on the floor, the Moon going round the Earth and the Earth going round the Sun. In this topic we describe gravity as a **field**, work out how strong it is at any distance from a planet, meet **gravitational potential**, and then use Newton's laws to explain how satellites, moons and planets orbit. Electric fields follow almost the same rules, so we finish by comparing the two.

Gravitational fields and field strength

Nothing has to touch you for the Earth to pull on you: a skydiver speeds up with nothing attached to them. Physicists describe this "action at a distance" with the idea of a **field**. Every mass fills the space around it with a gravitational field, and any other mass placed in that field feels a force.

DEFINITION

Gravitational field and field strength

A **gravitational field** is a region in which a mass experiences a force.

The **gravitational field strength** g at a point is the gravitational force per unit mass acting on a small test mass placed at that point.

Gravitational field strength

FORMULA SHEET

$$g = \frac{F}{m}$$

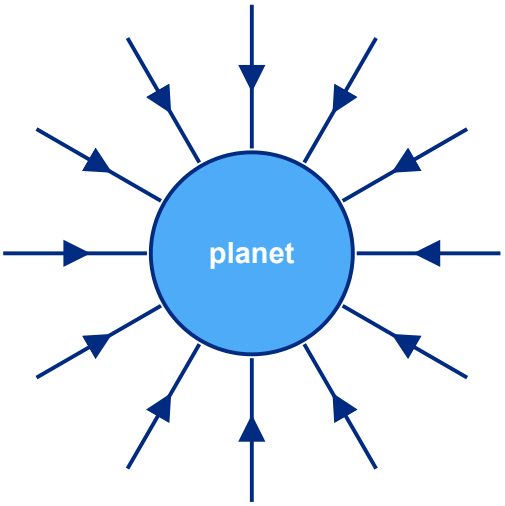
g: gravitational field strength / N kg^{-1} F: gravitational force on the mass / N m: mass placed in the field / kg

Field strength is a **vector**. It points the same way as the force on a mass, which is always **towards** the mass that creates the field. We imagine a **small** test mass so that its own field doesn't disturb the field we are trying to measure.

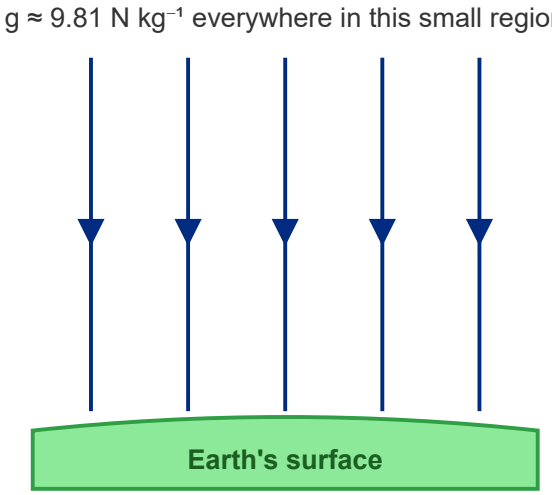
KEY IDEA

Field strength is the acceleration of free fall

The units N kg^{-1} and m s^{-2} are the same thing (1 N = 1 kg m s^{-2}). Newton's second law says $a = F/m$, which is exactly $g = F/m$. So "the field strength at the Earth's surface is 9.81 N kg^{-1}" and "objects fall with an acceleration of 9.81 m s^{-2}" say the same thing. Every object at the same place gets the same acceleration, whatever its mass, because doubling m doubles the force too.



radial field: lines spread out, field gets weaker



near the surface: almost uniform

Field lines show the direction of the force on a mass, and their spacing shows the strength. Around a planet the field is **radial**: the lines point in towards the centre and spread out with distance. Over a small region near the surface the lines are almost parallel and equally spaced, so the field is nearly **uniform**. That is why at AS you could use $g = 9.81\ \text{N kg}^{-1}$ and $\Delta E = mg\Delta h$.

EXAM TIP

When you draw gravitational field lines, always put arrows on them pointing **towards** the mass. Lines never cross, and they meet the surface of a spherical planet at right angles, heading for its centre.

Newton's law of universal gravitation

Newton's big idea (1687) was that the force that makes an apple fall is the same force that holds the Moon in its orbit. His law says that any two point masses attract each other with a force that is

- **proportional to the product of their masses**: double either mass and the force doubles;
- **inversely proportional to the square of the distance between their centres**: double the distance and the force drops to a quarter.

Newton's law of universal gravitation

FORMULA SHEET

$$F = \frac{Gm_1m_2}{r^2}$$

F: force on each mass / N m_1, m_2: the two masses / kg r: distance between their centres / m

$G = 6.67 \times 10^{-11}\ \text{N m}^2\,\text{kg}^{-2}$

DEFINITION

Newton's law of gravitation

Two point masses attract each other with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

KEY IDEA

Why the *square* of the distance?

Picture the field lines leaving a planet. At a distance r they are spread over the surface of a sphere of area $4\pi r^2$. Double the distance and the same number of lines is shared over four times the area, so the field, and the force on a mass, is a quarter as strong. Treble it and you get one ninth. Anything that spreads out evenly in three dimensions without being absorbed follows an inverse-square law. You will meet the same $4\pi r^2$ in electric fields and in the intensity of light from stars.

G is tiny, which is why you never notice the pull between everyday objects. Two 70 kg people standing 1 m apart attract each other with about 3×10^{-7} N, roughly the weight of a grain of sand. Gravity only matters when at least one of the masses is enormous, like a planet or a star.

The force obeys **Newton's third law**: the Earth pulls the Moon, and the Moon pulls the Earth with a force of exactly the same size in the opposite direction. The effects are very different only because the masses are.

COMMON MISTAKE

r is measured between the **centres** of the masses. For a satellite 400 km above the Earth, $r = 6.37 \times 10^6 + 4.0 \times 10^5 = 6.77 \times 10^6$ m, not 4.0×10^5 m. And don't forget to square it.

WORKED EXAMPLE

The Earth (mass 5.97×10^{24} kg) and the Moon (mass 7.35×10^{22} kg) are 3.84×10^8 m apart. Calculate the gravitational force between them and the acceleration each one has because of it.

Show solution

$$F = \frac{Gm_1m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 7.35 \times 10^{22}}{(3.84 \times 10^8)^2} = 1.98 \times 10^{20} \text{ N}$$

The same size of force acts on each body (Newton's third law), so use $a = F/m$ for each:

$$a_{\text{Moon}} = \frac{1.98 \times 10^{20}}{7.35 \times 10^{22}} = 2.70 \times 10^{-3} \text{ m s}^{-2}, \qquad a_{\text{Earth}} = \frac{1.98 \times 10^{20}}{5.97 \times 10^{24}} = 3.32 \times 10^{-5} \text{ m s}^{-2}$$

The Moon's acceleration is the centripetal acceleration that keeps it in orbit. The Earth's is about 80 times smaller: it makes the Earth wobble slightly as the Moon goes round.

The field around a point mass: g = Gm/r²

Now combine the two equations you have met so far. This is a derivation the specification says you must be able to do, and it only takes three lines.

Deriving g = Gm/r² for a point mass

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Place a small test mass m a distance r from a point mass M. Newton's law of gravitation gives the force on it:

$$F = \frac{GMm}{r^2}$$

2. Gravitational field strength is defined as force per unit mass, so divide by m:

$$g = \frac{F}{m} = \frac{GMm}{r^2} \cdot \frac{1}{m}$$

3. The test mass cancels:

$$\boxed{g = \frac{GM}{r^2}}$$

The field depends only on the mass that creates it and how far away you are, not on what you put there. That is why a hammer and a feather dropped together on the Moon land together.

4. The field points towards M. Outside a spherical planet or star, the field is the same as if all its mass were concentrated at its centre, so the equation works for any r from the surface outwards, with r measured from the centre.

Newton proved this "shell theorem" with some clever geometry. The proof is not in the spec; you just need to know you can treat a sphere as a point mass.

Radial gravitational field

FORMULA SHEET

$$g = \frac{Gm}{r^2}$$

g: field strength / N kg⁻¹ m: mass creating the field / kg r: distance from its centre / m

Because $g \propto 1/r^2$, going from the surface ($r = R$) to $r = 2R$ divides g by 4, and $r = 3R$ divides it by 9. One Earth radius above the ground, g is only about 2.45 N kg⁻¹.

WORKED EXAMPLE

The Earth's radius is 6.37×10^6 m and g at its surface is 9.81 N kg⁻¹. (a) Calculate the mass of the Earth. (b) The International Space Station orbits 408 km above the surface. Calculate g at that height.

Show solution

(a) Rearrange $g = GM/R^2$:

$$M = \frac{gR^2}{G} = \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.97 \times 10^{24} \text{ kg}$$

(b) Measure from the centre: $r = 6.37 \times 10^6 + 0.408 \times 10^6 = 6.778 \times 10^6$ m.

$$g = \frac{GM}{r^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{(6.778 \times 10^6)^2} = 8.67 \text{ N kg}^{-1}$$

That is still 88% of the value on the ground. Gravity has certainly not "switched off" up there, which matters when we explain why astronauts float.

Two masses: where do the fields cancel?

Field strengths are vectors, so where two fields overlap you add them as vectors. On the line between the Earth and the Moon the two pulls point in opposite directions. Close to the Earth its pull wins easily, but somewhere near the Moon the two are equal and opposite and the resultant field is zero. This is the **neutral point**.

WORKED EXAMPLE

Using the data from the first worked example, find how far from the centre of the Earth the resultant gravitational field on the Earth–Moon line is zero.

Show solution

Let the point be a distance x from the Earth, so $d - x$ from the Moon, with $d = 3.84 \times 10^8$ m. The two field strengths are equal there:

$$\frac{GM_E}{x^2} = \frac{GM_M}{(d-x)^2} \quad \Rightarrow \quad \frac{d-x}{x} = \sqrt{\frac{M_M}{M_E}} = \sqrt{\frac{7.35 \times 10^{22}}{5.97 \times 10^{24}}} = 0.111$$

Taking the square root of both sides keeps the algebra linear. Use the positive root: the point is between the two bodies.

$$d = 1.111\,x \quad \Rightarrow \quad x = \frac{3.84 \times 10^8}{1.111} = 3.46 \times 10^8 \text{ m}$$

That is 90% of the way to the Moon, about 3.8×10^7 m from the Moon's centre. G cancelled, so you didn't need its value.

Gravitational potential

Lift a book and you do work against gravity; let go and the field does work on it as it falls. **Gravitational potential** tells you how much work per kilogram is involved in getting to each point in a field. It works like height on a map: the field pulls masses "downhill" towards lower potential.

DEFINITION

Gravitational potential

The gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point. It is a **scalar**, measured in J kg⁻¹.

Gravitational potential in a radial field

FORMULA SHEET

$$V_{\text{grav}} = -\frac{Gm}{r}$$

V_{grav}: potential / J kg⁻¹ m: mass creating the field / kg r: distance from its centre / m

KEY IDEA

Why is the potential negative?

We choose the potential to be **zero at infinity**, far from every mass. Gravity is attractive, so as a mass falls in from infinity the field does the work; to bring it in slowly you would have to hold it back, which means *you* do negative work. So at every finite distance $V < 0$, and the closer you get to the mass the more negative it becomes. A mass near a planet sits at the bottom of a "potential well", and you must supply energy to lift it out.

Notice that V goes as $1/r$, not $1/r^2$: the potential fades away more slowly than the field strength. At the Earth's surface $V = -6.25 \times 10^7$ J kg^{-1} and at twice that distance it is half as big, while g has dropped to a quarter.

Potential difference and potential energy

To move a mass m between two points you must do work equal to the mass times the change in potential: $\Delta W = m\,\Delta V$. It follows that a mass m at a point where the potential is V_{grav} has gravitational potential energy

$$E_{\text{grav}} = mV_{\text{grav}} = -\frac{GMm}{r}$$

This is also negative, and zero at infinity. Close to the Earth, over a height change that is small compared with the radius, $\Delta V \approx g\,\Delta h$, which gives back the AS result $\Delta E = mg\Delta h$. Over large distances g changes, so you must use ΔV instead.

WORKED EXAMPLE

A 1000 kg satellite is to be placed in a circular orbit 400 km above the Earth (radius 6.37×10^6 m, mass 5.97×10^{24} kg). (a) Calculate the increase in its gravitational potential energy. (b) Compare your answer with $mg\Delta h$. (c) Calculate the kinetic energy it needs in orbit.

Show solution

(a) $GM = 6.67 \times 10^{-11} \times 5.97 \times 10^{24} = 3.982 \times 10^{14}$ N m^2 kg^{-1}. Find the potential at each radius:

$$V_1 = -\frac{3.982 \times 10^{14}}{6.37 \times 10^6} = -6.251 \times 10^7 \text{ J kg}^{-1}, \qquad V_2 = -\frac{3.982 \times 10^{14}}{6.77 \times 10^6} = -5.882 \times 10^7 \text{ J kg}^{-1}$$

$$\Delta E = m\,\Delta V = 1000 \times (-5.882 \times 10^7 - (-6.251 \times 10^7)) = 3.69 \times 10^9 \text{ J}$$

(b) $mg\Delta h = 1000 \times 9.81 \times 4.0 \times 10^5 = 3.92 \times 10^9$ J. This is about 6% too big, because g gets smaller as the satellite rises (it is 8.69 N kg⁻¹ at 400 km).

(c) In a circular orbit $v^2 = GM/r$ (see the orbits section below), so

$$E_k = \tfrac{1}{2}mv^2 = \frac{GMm}{2r} = \frac{3.98 \times 10^{14} \times 1000}{2 \times 6.77 \times 10^6} = 2.94 \times 10^{10} \text{ J}$$

Eight times the potential energy. Getting up to orbit is the easy part; getting fast enough to stay there is what needs the huge rockets.

Equipotentials and field lines

An **equipotential** is a surface (a line on a flat diagram) on which the potential is the same everywhere. Around a single planet they are spheres. Three things to know:

- No work is done moving a mass along an equipotential, because $\Delta V = 0$.
- Equipotentials are always **perpendicular to field lines**. If they weren't, the field would have a component along the equipotential and would do work there.
- If you draw them at equal steps of ΔV, they crowd together where the field is strong, like contour lines on a steep hill.

Field strength is the potential gradient

The field strength tells you how steeply the potential changes with distance:

$$g = -\frac{\Delta V}{\Delta r}$$

On a graph of V against r, the gradient at any point equals the size of g there. The minus sign says the field points the way the potential *decreases*, which is towards the mass, just as a ball rolls downhill. You can check the units: J kg⁻¹ m⁻¹ = N m kg⁻¹ m⁻¹ = N kg⁻¹. The same link between field and potential holds for electric fields.

Orbital motion

Newton imagined a cannon on top of a very high mountain, firing balls horizontally. A slow ball curves down and hits the ground nearby. A faster one lands further away. Fire it fast enough and, as it falls, the curved Earth drops away beneath it at the same rate: it keeps falling but never lands. That is an orbit. **An orbiting satellite is in continuous free fall.**

KEY IDEA

Newton's three laws in an orbit

- **First law:** without a force the satellite would fly off in a straight line along the tangent. Something must keep pulling it round.
- **Second law:** that something is gravity, the only force acting. It is the resultant force, so it causes an acceleration towards the centre: it provides the **centripetal force** mv^2/r (see Circular Motion).

- **Third law:** the satellite pulls the Earth with an equal and opposite force. Because the Earth is so massive, its acceleration is negligible.

Orbital speed of a satellite

1. For a satellite of mass m in a circular orbit of radius r round a planet of mass M, gravity provides the centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

2. Cancel m and one factor of r:

$$v^2 = \frac{GM}{r}$$

3. Take the square root:

$$\boxed{v = \sqrt{\frac{GM}{r}}}$$

The satellite's mass has cancelled, so a spanner and a space station in the same orbit move at the same speed. A bigger r means a smaller v: higher orbits are slower.

EXAM TIP

Why do astronauts float?

Not because there is no gravity: g at the ISS is 8.7 N kg^{-1}. The astronauts and the station are falling together with the same acceleration, so the floor doesn't need to push on them, and they feel weightless. Examiners like the phrase "in free fall, with the same acceleration as the spacecraft".

Period and radius: T² ∝ r³

A satellite goes once round its orbit, a distance $2\pi r$, in one period T, so $v = 2\pi r/T$. Put that into the orbital speed equation and you get a relationship between the period and the radius of any orbit.

Period of a circular orbit

1. Start from $v^2 = GM/r$ and substitute $v = 2\pi r/T$:

$$\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r}$$

2 Rearrange to make T^2 the subject:

$$\boxed{T^2 = \frac{4\pi^2}{GM} r^3}$$

Kepler discovered $T^2 \propto r^3$ from planetary observations in 1619 ("Kepler's third law"), almost 70 years before Newton's law of gravitation explained it.

3 So for everything orbiting the same central body, a graph of T^2 against r^3 is a straight line through the origin with

$$\text{gradient} = \frac{4\pi^2}{GM} \quad \Rightarrow \quad M = \frac{4\pi^2}{G \times \text{gradient}}$$

This is how astronomers "weigh" planets and stars: time how long a moon takes to go round and measure the size of its orbit. You don't need to know anything about the moon itself.

WORKED EXAMPLE

Io orbits Jupiter with a period of 1.77 days at a radius of 4.22×10^8 m. Calculate the mass of Jupiter.

Show solution

Convert the period to seconds: $T = 1.77 \times 24 \times 3600 = 1.529 \times 10^5$ s. Rearrange $T^2 = 4\pi^2 r^3 / GM$:

$$M = \frac{4\pi^2 r^3}{GT^2} = \frac{4\pi^2 \times (4.22 \times 10^8)^3}{6.67 \times 10^{-11} \times (1.529 \times 10^5)^2} = 1.90 \times 10^{27} \text{ kg}$$

About 320 times the mass of the Earth.

Geostationary orbits

A **geostationary** satellite stays above the same point on the Earth's surface. For that it must

- have a period of **24 hours**, the same as the Earth's rotation;
- orbit above the **equator**;
- move in the **same direction** as the Earth rotates (west to east).

Because T is fixed, $T^2 \propto r^3$ fixes the radius too: there is only one geostationary orbit. Communications and TV satellites use it, so a dish on your roof can point at one spot in the sky and never move. Low orbits like the ISS (about 90 minutes per orbit) are better for photographing the ground in detail, and GPS satellites take about 12 hours.

WORKED EXAMPLE

Calculate the height above the Earth's surface of a geostationary orbit, and the speed of a satellite in it. Mass of Earth $= 5.97 \times 10^{24}$ kg, radius of Earth $= 6.37 \times 10^6$ m.

Show solution

$T = 24 \times 3600 = 86\,400$ s. Rearrange $T^2 = 4\pi^2 r^3 / GM$ for r:

$$r^3 = \frac{GMT^2}{4\pi^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times (86\,400)^2}{4\pi^2} = 7.53 \times 10^{22} \text{ m}^3$$

$$r = \sqrt[3]{7.53 \times 10^{22}} = 4.22 \times 10^7 \text{ m}$$

This is measured from the centre, so subtract the Earth's radius:

$$h = 4.22 \times 10^7 - 6.37 \times 10^6 = 3.59 \times 10^7 \text{ m} \quad (\text{about } 36\,000 \text{ km})$$

$$v = \frac{2\pi r}{T} = \frac{2\pi \times 4.22 \times 10^7}{86\,400} = 3.07 \times 10^3 \text{ m s}^{-1}$$

COMMON MISTAKE

Two classic slips in orbit questions: leaving T in hours or days (it must be in **seconds**), and giving r as the answer when the question asks for the **height above the surface**.

Comparing gravitational and electric fields

Gravitational fields act on **mass**; electric fields act on **charge**. Apart from that, the maths is almost identical, so learning one helps you with the other (see Electric Fields).

	Gravitational field	Electric field
Acts on	mass	charge
Field strength	$g = F/m$, N kg⁻¹	$E = F/Q$, N C⁻¹
Force between two point objects	$F = \frac{Gm_1m_2}{r^2}$	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2}$
Field of a point object	$g = \frac{Gm}{r^2}$	$E = \frac{Q}{4\pi\varepsilon_0 r^2}$
Potential in a radial field	$V = -\frac{Gm}{r}$: always negative	$V = \frac{Q}{4\pi\varepsilon_0 r}$: positive or negative
Attract or repel?	always attracts	like charges repel, unlike attract
Constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}	$\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9$ N m^2 C^{-2}

Uniform field	near a planet's surface (approximately)	between parallel plates, $E = V/d$

KEY IDEA

Similarities and differences

Similar: both follow an inverse-square law for a point source; both radial fields look the same on a diagram; both have equipotentials perpendicular to the field lines; in both, the potential is zero at infinity and the field strength is the potential gradient.

Different: gravity only ever attracts, because there is no negative mass, so gravitational potential is never positive. Electric forces can attract or repel. And gravity is extraordinarily weak by comparison: it only dominates on large scales because big objects are almost exactly neutral, so their electric forces cancel while their gravity adds up.

WORKED EXAMPLE

In a hydrogen atom an electron and a proton attract each other electrically and gravitationally. Show that the electric force is about 10^{39} times bigger. ($e = 1.60 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg, $m_e = 9.11 \times 10^{-31}$ kg)

Show solution

Both forces have r^2 underneath, so it cancels when you divide one by the other:

$$\frac{F_E}{F_G} = \frac{e^2/4\pi\varepsilon_0 r^2}{Gm_p m_e/r^2} = \frac{8.99 \times 10^9 \times (1.60 \times 10^{-19})^2}{6.67 \times 10^{-11} \times 1.67 \times 10^{-27} \times 9.11 \times 10^{-31}} = 2.27 \times 10^{39}$$

This ratio doesn't depend on how far apart they are, because both are inverse-square laws.

Quick checks

QUICK CHECK

The gravitational field strength at the Earth's surface is 9.81 N kg⁻¹. What is it at a height above the surface equal to the Earth's radius?

- 4.91 N kg⁻¹
- 2.45 N kg⁻¹
- 1.09 N kg⁻¹
- 9.81 N kg⁻¹

The distance from the **centre** doubles (from R to $2R$), so g falls by a factor of $2^2 = 4$: $9.81/4 = 2.45\ \text{N kg}^{-1}$. 4.91 forgets the square; 1.09 uses $3R$.

QUICK CHECK

A satellite is moved from a low circular orbit to a higher circular orbit around the Earth. What happens to its speed and its period?

- Speed increases, period increases
- Speed increases, period decreases
- Speed decreases, period increases
- Speed stays the same, period increases

$v = \sqrt{GM/r}$, so a bigger r means a lower speed. It also has further to go, so $T = 2\pi r/v$ increases on both counts ($T^2 \propto r^3$).

QUICK CHECK

Why do astronauts on the International Space Station feel weightless?

- The gravitational field strength at that height is zero.
- They are outside the Earth's atmosphere.
- The Moon's gravity cancels the Earth's gravity.
- They and the station are in free fall with the same acceleration.

g at the ISS is about 8.7 N kg^{-1}, nearly the surface value. Everything on board is falling round the Earth together, so nothing presses on anything else.

QUICK CHECK

At a point in the Earth's field the gravitational potential is −50 MJ kg^{-1}. What is the minimum energy needed to move a 2.0 kg mass from there to infinity?

- 25 MJ
- 50 MJ
- 100 MJ
- −100 MJ

The potential at infinity is zero, so $\Delta V = 0 - (-50) = +50\ \text{MJ kg}^{-1}$. Work done $= m\Delta V = 2.0 \times 50 = 100\ \text{MJ}$. You have to put energy in, so it is positive.

QUICK CHECK

At the neutral point between the Earth and the Moon, which statement is correct?

- Both the field strength and the potential are zero.
- The field strength is zero but the potential is negative.
- The potential is zero but the field strength is not.
- The field strength is at a maximum.

Field strengths are vectors and cancel there. Potentials are scalars and simply add: both are negative, so the total is negative. The potential has a maximum at that point, so its gradient (the field) is zero.

QUICK CHECK

Moon X orbits a planet at four times the orbital radius of moon Y. How many times longer is the period of X than the period of Y?

- 2
- 4
- 8
- 16

$T^2 \propto r^3$, so $T \propto r^{3/2}$. Multiplying r by 4 multiplies T by $4^{3/2} = \sqrt{64} = 8$.

A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Astrophysics & Cosmology

We will never visit another star, yet we know how hot stars are, how big and how far away they are, and how they live and die. All of it comes from light that reaches us: a star's spectrum gives its temperature, its brightness and distance give its power and size, and the redshift of distant galaxies reveals an expanding universe with a finite age.

Black-body radiation

Everything emits electromagnetic radiation because of its temperature. You can't see your own glow because at 37 °C it is all infrared, but heat a poker and it glows dull red, then orange, then yellow-white. The hotter an object is, the **more** it radiates and the **shorter** the wavelengths it radiates most strongly.

DEFINITION

Black body

A black body is an object that absorbs **all** the electromagnetic radiation that falls on it. It is also the best possible emitter, and the spectrum it emits depends only on its **temperature**.

A black body only looks black when it is cold. Stars are very close to black bodies: their hot, dense gas absorbs any radiation falling on it, and the light they give out depends almost entirely on their surface temperature. That is how we read a star's temperature from its light.

Reading a black-body radiation curve

A radiation curve plots the intensity emitted at each wavelength. Every black-body curve has the same shape:

- It is a **continuous** spectrum: some radiation at every wavelength, with a single peak at $\lambda_{\max}$.
- It rises steeply on the short-wavelength side and has a long tail on the long-wavelength side.
- A **hotter** body has a curve that is higher at **every** wavelength, so the curves for different temperatures never cross.
- A hotter body has its peak at a **shorter** wavelength (Wien's law), and a much bigger **area** under the curve, which represents the total power emitted per square metre (Stefan–Boltzmann law).

EXAM TIP

When you sketch black-body curves for two temperatures, make the hotter curve higher at **every** wavelength and put its peak further to the **left** (shorter wavelength). Students often draw the curves crossing, which loses the mark.

Luminosity and temperature

DEFINITION

Luminosity

The luminosity L of a star is the total power it emits as electromagnetic radiation, at all wavelengths and in all directions. It is measured in watts. The Sun's luminosity is 3.83×10^{26} W.

The Stefan–Boltzmann law

Stefan–Boltzmann law

FORMULA SHEET

$$L = \sigma A T^4$$

L: luminosity / W $\sigma = 5.67 \times 10^{-8}\ \text{W m}^{-2}\,\text{K}^{-4}$ A: surface area / m² ($4\pi r^2$ for a star)
T: surface temperature / K

σT^4 is the power radiated by each square metre of surface, and A tells you how many square metres there are. Because of the fourth power, temperature matters enormously: double the temperature and each square metre gives out $2^4 = 16$ times as much power. Since $A = 4\pi r^2$ for a sphere, $L \propto r^2 T^4$: a star can be very luminous by being hot, by being huge, or both.

COMMON MISTAKE

Temperatures must be in **kelvin**, A is the surface area $4\pi r^2$ (not πr^2, and r is the radius, not the diameter), and the fourth power applies to T only. On a calculator, type T then the power key, then 4.

Wien's law

Wien's law

FORMULA SHEET

$$\lambda_{\text{max}}\, T = 2.898 \times 10^{-3}\ \text{m K}$$

λ_{max}: wavelength at the peak of the curve / m T: surface temperature / K

The peak wavelength is inversely proportional to the temperature: double T and $\lambda_{\max}$ halves. Astronomers measure the spectrum of a star, find where the peak is, and get the surface temperature without going anywhere near it. This is why **colour tells you temperature**: red stars are cool (about 3000 K), yellow-white stars like the Sun are about 6000 K, and blue-white stars are 10 000 K or more.

WORKED EXAMPLE

The Sun has luminosity 3.83×10^{26} W and radius 6.96×10^{8} m. Calculate its surface temperature and the wavelength at which its spectrum peaks.

Show solution

Rearrange $L = \sigma 4\pi r^2 T^4$:

$$T^4 = \frac{L}{4\pi r^2 \sigma} = \frac{3.83 \times 10^{26}}{4\pi \times (6.96 \times 10^{8})^2 \times 5.67 \times 10^{-8}} = 1.11 \times 10^{15}\ \text{K}^4$$

$$T = (1.11 \times 10^{15})^{1/4} = 5770\ \text{K}$$

Then Wien's law:

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}}{5770} = 5.02 \times 10^{-7}\ \text{m} = 502\ \text{nm}$$

That is green light, in the middle of the visible spectrum. The Sun gives out plenty of every visible colour, which is why sunlight looks white rather than green.

WORKED EXAMPLE

A red supergiant has a luminosity of 4.5×10^{31} W and its spectrum peaks at 8.5×10^{-7} m. Calculate its radius and compare it with the Sun's (6.96×10^{8} m).

Show solution

First the temperature, from Wien's law:

$$T = \frac{2.898 \times 10^{-3}}{8.5 \times 10^{-7}} = 3410\ \text{K}$$

Then rearrange $L = 4\pi r^2 \sigma T^4$ for r:

$$r = \sqrt{\frac{L}{4\pi\sigma T^4}} = \sqrt{\frac{4.5 \times 10^{31}}{4\pi \times 5.67 \times 10^{-8} \times 3410^4}} = 6.8 \times 10^{11}\ \text{m}$$

That is about 980 times the Sun's radius. Placed where the Sun is, it would reach most of the way to Jupiter's orbit. It is cooler than the Sun but 120 000 times more luminous because its surface area is a million times bigger.

Intensity and the inverse-square law

How bright a star looks depends on how much power it emits and how far away it is. As the light travels out, the same power is spread over a bigger and bigger sphere.

Intensity at a distance d from a star

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A star of luminosity L radiates equally in all directions. At a distance d, all of that power passes through the surface of a sphere of radius d:

$$A = 4\pi d^2$$

2. Intensity is power per unit area, measured in W m⁻²:

$$\boxed{I = \frac{L}{4\pi d^2}}$$

Double the distance and the area is four times bigger, so the intensity is a quarter: the same inverse-square law as gravity.

Intensity of radiation

FORMULA SHEET

$$I = \frac{L}{4\pi d^2}$$

I: intensity received / W m⁻² L: luminosity of the source / W d: distance from the source / m

For example, the Earth is 1.50×10^{11} m from the Sun, so sunlight arriving at the top of the atmosphere has intensity $I = 3.83 \times 10^{26}/(4\pi \times (1.50 \times 10^{11})^2) = 1.35 \times 10^3\ \text{W m}^{-2}$. Measuring I is easy with a telescope

and a detector. So if you know any two of I, L and d, you can find the third.

Measuring distances to stars and galaxies

Trigonometric parallax

Hold a finger up at arm's length and look at it with one eye, then the other. It jumps sideways against the background. A nearby star does the same as the Earth moves round the Sun: observed six months apart, from opposite sides of the Earth's orbit, it shifts slightly against the much more distant stars. The closer the star, the bigger the shift.

DEFINITION

Parallax angle and the parsec

The **parallax angle** p is the angle subtended at the star by the radius of the Earth's orbit (1 AU). It is half the total angular shift of the star seen six months apart.

One **parsec** (pc) is the distance at which 1 AU subtends an angle of 1 arcsecond (1/3600 of a degree). 1 pc = 3.09×10^{16} m = 3.26 light-years.

Distance from the parallax angle

1. The Sun, the Earth and the star make a right-angled triangle. The side opposite p is 1 AU, and the long side is the distance d:

$$\tan p = \frac{1\ \text{AU}}{d}$$

2. The angle is tiny, so $\tan p \approx p$ with p in radians:

$$d = \frac{1\ \text{AU}}{p}$$

3. 1 arcsecond is $\frac{1}{3600} \times \frac{\pi}{180} = 4.85 \times 10^{-6}$ rad, so a star with p = 1 arcsecond is at $1.50 \times 10^{11} / 4.85 \times 10^{-6} = 3.09 \times 10^{16}$ m, which is 1 pc. In these units:

$$\boxed{d\ (\text{pc}) = \frac{1}{p\ (\text{arcseconds})}}$$

WORKED EXAMPLE

Proxima Centauri, the nearest star to the Sun, has a parallax angle of 0.768 arcseconds. Calculate its distance in parsecs, metres and light-years. (1 light-year = 9.46×10^{15} m)

Show solution

$$d = \frac{1}{0.768} = 1.30 \text{ pc}$$

$$d = 1.30 \times 3.09 \times 10^{16} = 4.02 \times 10^{16} \text{ m} = \frac{4.02 \times 10^{16}}{9.46 \times 10^{15}} = 4.25 \text{ light-years}$$

Parallax angles are tiny and the atmosphere blurs star images, so from the ground the method only works out to about 100 pc. Space telescopes such as Gaia reach thousands of parsecs, still only part of our own galaxy (about 30 000 pc across). Further away we need another method.

Standard candles

DEFINITION

Standard candle

A standard candle is an astronomical object of **known luminosity**.

If you know how much power an object emits, measure the intensity I that arrives at Earth and rearrange $I = L/4\pi d^2$:

$$d = \sqrt{\frac{L}{4\pi I}}$$

It is the same idea as judging how far away a car is at night from how bright its headlights look, because you know roughly how bright headlights are. Two important standard candles are **Cepheid variable stars**, which pulse in brightness, and **type Ia supernovae**, exploding white dwarfs so bright that they can be seen in galaxies billions of light-years away.

WORKED EXAMPLE

The star Sirius A has a parallax angle of 0.379 arcseconds. The intensity of its radiation at the Earth is 1.17×10^{-7} W m^{-2} and its spectrum peaks at 292 nm. Calculate (a) its distance, (b) its luminosity, (c) its surface temperature and (d) its radius.

Show solution

(a) $d = 1/0.379 = 2.64 \text{ pc} = 2.64 \times 3.09 \times 10^{16} = 8.15 \times 10^{16} \text{ m}$

(b) Rearrange $I = L/4\pi d^2$:

$$L = 4\pi d^2 I = 4\pi \times (8.15 \times 10^{16})^2 \times 1.17 \times 10^{-7} = 9.77 \times 10^{27} \text{ W}$$

(c) Wien's law:

$$T = \frac{2.898 \times 10^{-3}}{292 \times 10^{-9}} = 9920 \text{ K}$$

(d) Stefan–Boltzmann law:

$$r = \sqrt{\frac{L}{4\pi\sigma T^4}} = \sqrt{\frac{9.77 \times 10^{27}}{4\pi \times 5.67 \times 10^{-8} \times 9920^4}} = 1.19 \times 10^9 \text{ m}$$

Sirius A is about 1.7 times the radius of the Sun and 25 times as luminous. Notice the chain: parallax gives d, d and I give L, the spectrum gives T, and L and T give r.

EXAM TIP

In these multi-step questions, convert parsecs or light-years to **metres** before using $I = L/4\pi d^2$, and nanometres to metres before using Wien's law.

The Hertzsprung–Russell diagram

Around 1910 Ejnar Hertzsprung and Henry Russell plotted the luminosity of stars against their surface temperature. Instead of a random scatter, the stars fell into a few distinct groups.

- **Luminosity** goes up the vertical axis, usually in multiples of the Sun's luminosity, on a logarithmic scale (each step is ten times the last).
- **Surface temperature** goes along the horizontal axis, also on a log scale, and it **decreases** from left to right: hot blue stars on the left, cool red stars on the right.

Region	Where on the diagram	What the stars are like
Main sequence	diagonal band from top left to bottom right	about 90% of stars, including the Sun; fusing hydrogen into helium in their cores. Hotter ones are more massive and more luminous.

Red giants	above the main sequence, on the right	cool (3000–5000 K) but 10 to 1000 times more luminous than the Sun, so they must be large
Supergiants	across the very top	the most luminous stars, up to a million times the Sun; enormous
White dwarfs	bottom left	hot but very dim, so they must be tiny: about the size of the Earth

KEY IDEA

Position on the diagram tells you the size

From $L = 4\pi r^2 \sigma T^4$, two stars with the same temperature but different luminosities must have different radii: $r \propto \sqrt{L}$ at fixed T. So stars at the top right are cool and very luminous, which means huge; stars at the bottom left are hot and dim, which means tiny. Lines of constant radius run diagonally across the diagram.

EXAM TIP

To **sketch** an HR diagram: label the axes "luminosity / $L_\odot$" (or in W) and "surface temperature / K", put the temperature scale the right way round (hot on the left, for example 40 000 K down to 3000 K), mark the main sequence as a band from top left to bottom right, add the red giants, supergiants and white dwarfs, and mark the Sun at about 6000 K and 1 $L_\odot$.

WORKED EXAMPLE

Sirius B is a white dwarf with surface temperature 25 000 K and luminosity 2.1×10^{25} W. Show that it is only a little larger than the Earth (radius 6.37×10^6 m).

Show solution

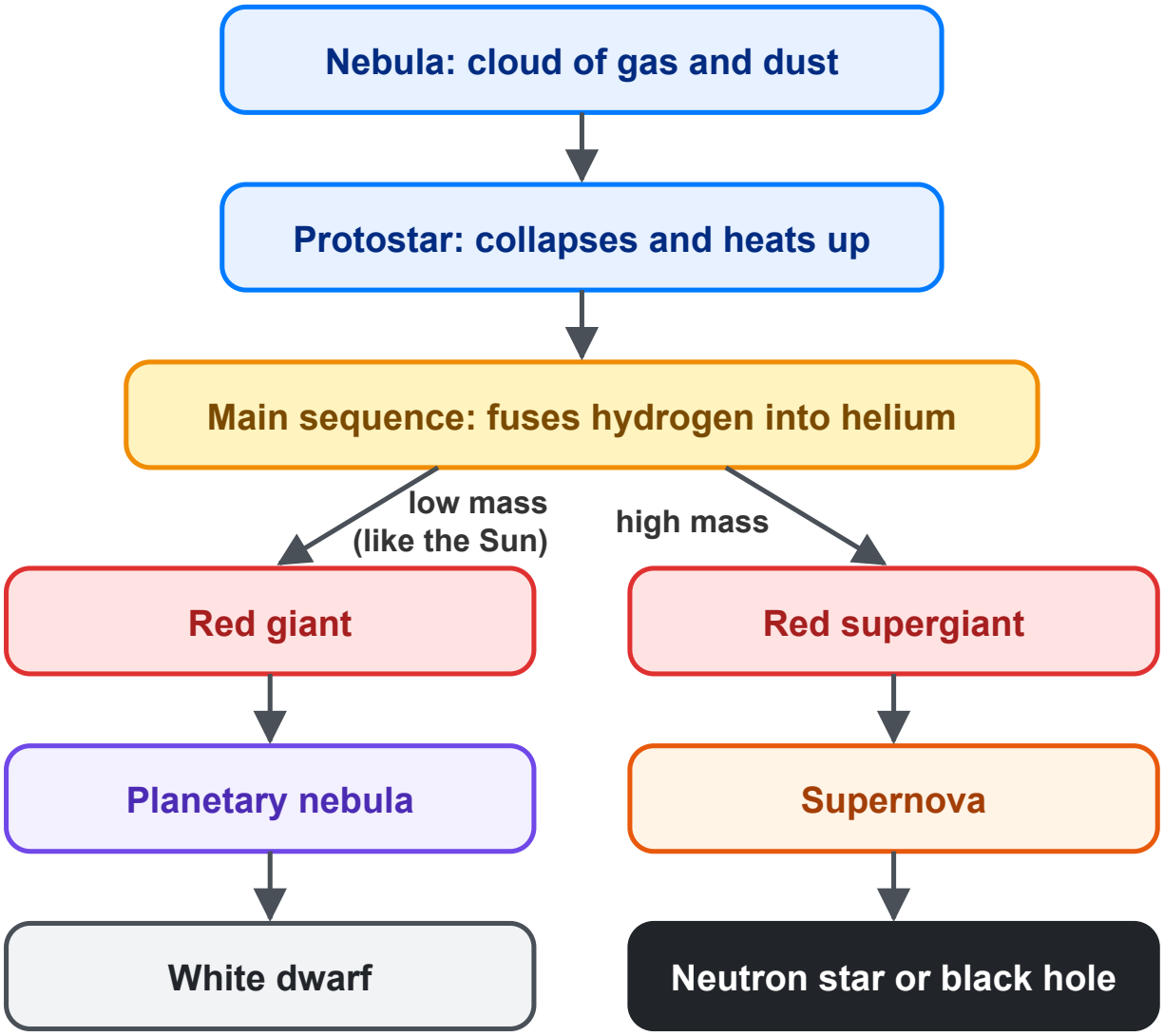
$$r = \sqrt{\frac{L}{4\pi\sigma T^4}} = \sqrt{\frac{2.1 \times 10^{25}}{4\pi \times 5.67 \times 10^{-8} \times (25\,000)^4}} = 8.7 \times 10^6 \text{ m}$$

That is about 1.4 Earth radii, yet Sirius B has about the same mass as the Sun. It is hotter than the Sun but 18 times less luminous, because its surface area is so small.

The life cycle of stars

A star's whole life is a balance between **gravity**, pulling everything inwards, and **pressure** from the hot gas and radiation, pushing outwards. Which stages it goes through depends mainly on its **mass**.

1. **Nebula to protostar.** A cloud of gas and dust, mostly hydrogen, collapses under its own gravity. Gravitational potential energy is converted into thermal energy and the core heats up.
2. **Main sequence.** When the core reaches about 10^7 K, hydrogen nuclei fuse into helium. The energy released keeps the core hot and the outward pressure balances gravity, so the star is stable for most of its life. More massive stars are hotter and more luminous and use up their hydrogen much faster: about 10 billion years for the Sun, only about 10 million years for a star of 15 solar masses.
3. **Giant stage.** When the hydrogen in the core runs out, the core contracts and heats up, hydrogen fuses in a shell around it, and the outer layers swell up and cool. The star becomes a **red giant** (or, if it is massive, a **red supergiant**): it moves up and to the right on the HR diagram. The core gets hot enough to fuse helium into carbon and oxygen.
4. **Low-mass stars, like the Sun:** the outer layers drift away as a glowing **planetary nebula**, leaving the hot core as a **white dwarf**. No fusion happens in it, so it slowly cools and fades, moving down and to the right on the HR diagram.
5. **High-mass stars:** fusion continues, making heavier and heavier elements up to iron. Fusing iron releases no energy, so when the core is iron it suddenly collapses, and the outer layers are blasted away in a **supernova**. The core that remains becomes a **neutron star**, about 20 km across and as dense as an atomic nucleus, or, for the most massive stars, a **black hole**, whose gravity is so strong that not even light can escape.



The two main paths of stellar evolution. Mass decides which path a star takes.

The Doppler effect and redshift

An ambulance siren sounds higher as it comes towards you and lower as it moves away. The waves themselves travel at the same speed, but the source moves between emitting one crest and the next. Ahead of the source the crests bunch up (shorter wavelength, higher frequency); behind it they are stretched out (longer wavelength, lower frequency). This is the **Doppler effect**, and it happens for any wave, including light.

KEY IDEA

Redshift and blueshift

Light from a source moving **away** from us has its wavelengths stretched towards the red end of the spectrum: a **redshift**. A source moving **towards** us gives a **blueshift**. We can measure the shift very precisely because star and galaxy spectra contain **absorption lines** at wavelengths we know from the

laboratory, like the hydrogen line at 656.3 nm. Compare where the lines are with where they should be and you get the speed.

Redshift for a slowly moving source

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 A source moving away at speed v emits a crest, then the next crest one period $T = 1/f$ later. In that time the first crest has travelled cT and the source has moved vT further away:

$$\lambda_{\text{observed}} = cT + vT = \lambda + vT$$

2 So the change in wavelength is $\Delta\lambda = vT$. Divide by $\lambda = cT$:

$$\frac{\Delta\lambda}{\lambda} = \frac{vT}{cT} = \frac{v}{c}$$

3 This fractional change is called the redshift, z. For speeds much less than c the fractional change in frequency is the same size, so

$$\boxed{z = \frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}}$$

For a redshift the frequency goes down, so Δf is negative; the equation uses its size.

The approximation only works when $v \ll c$.

Redshift of electromagnetic radiation

FORMULA SHEET

$$z = \frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$

z: redshift (no unit) $\Delta\lambda$: observed minus emitted wavelength / m

λ: emitted (laboratory) wavelength / m v: speed of the source along the line of sight / m s^{-1}

WORKED EXAMPLE

The hydrogen line that is at 656.3 nm in the laboratory is observed at 671.9 nm in the spectrum of a distant galaxy. Calculate the redshift and the galaxy's speed. Is it moving towards us or away from us?

Show solution

$$z = \frac{\Delta\lambda}{\lambda} = \frac{671.9 - 656.3}{656.3} = \frac{15.6}{656.3} = 0.0238$$

$$v = zc = 0.0238 \times 3.00 \times 10^{8} = 7.13 \times 10^{6}\ \text{m s}^{-1}$$

The wavelength has increased (redshift), so the galaxy is moving **away** from us, at about 2.4% of the speed of light.

Hubble's law and the age of the universe

In 1929 Edwin Hubble combined distances (from Cepheids) with redshifts for nearby galaxies. Almost every galaxy is **moving away** from us, and the further away it is, the faster it recedes. The recession velocity is proportional to distance:

Hubble's law (cosmological expansion)

FORMULA SHEET

$$v = H_0 d$$

v: recession velocity / m s⁻¹ or km s⁻¹ d: distance / m or Mpc

H_0: the Hubble constant, about 70 km s⁻¹ Mpc⁻¹ $= 2.3 \times 10^{-18}\ \text{s}^{-1}$

KEY IDEA

Why $v \propto d$ means the universe is expanding

Picture a loaf of raisin bread rising in the oven. Every raisin moves away from every other one, and a raisin twice as far from you moves away twice as fast, because there is twice as much dough between you to stretch. Any raisin would see exactly the same thing, so we are not at the centre; there is no centre. Run the film backwards and everything was once much closer together: the evidence for the universe beginning in a hot, dense **Big Bang**.

Astronomers usually quote H_0 in km s⁻¹ Mpc⁻¹ (1 Mpc = 10^6 pc = 3.09×10^{22} m). To convert to SI, write both parts in metres and seconds: $70\ \text{km s}^{-1}\,\text{Mpc}^{-1} = 7.0 \times 10^{4}\ \text{m s}^{-1} \div 3.09 \times 10^{22}\ \text{m} = 2.27 \times 10^{-18}\ \text{s}^{-1}$.

Estimating the age of the universe

1 Suppose a galaxy now at distance d has been moving away from us at a constant speed v since the Big Bang. The time it has taken is

$$t = \frac{d}{v}$$

2 Hubble's law says $v = H_0 d$, so d cancels and the answer is the same for every galaxy:

$$\boxed{t \approx \frac{1}{H_0}}$$

3 With $H_0 = 2.27 \times 10^{-18}\ \text{s}^{-1}$ and 1 year = 3.15×10^{7} s:

$$t = \frac{1}{2.27 \times 10^{-18}} = 4.4 \times 10^{17}\ \text{s} = 1.4 \times 10^{10}\ \text{years}$$

The estimate assumes the expansion rate has always been the same. If gravity has slowed the expansion, galaxies used to move faster and the true age is less than $1/H_0$.

WORKED EXAMPLE

A type Ia supernova in the galaxy from the redshift example has a peak luminosity of 2.0×10^{36} W, and its peak intensity at the Earth is $1.6 \times 10^{-14}\ \text{W m}^{-2}$. Use this and the galaxy's speed, $7.13 \times 10^{6}\ \text{m s}^{-1}$, to find the Hubble constant and estimate the age of the universe.

Show solution

The supernova is a standard candle, so rearrange $I = L/4\pi d^2$:

$$d = \sqrt{\frac{L}{4\pi I}} = \sqrt{\frac{2.0 \times 10^{36}}{4\pi \times 1.6 \times 10^{-14}}} = 3.15 \times 10^{24}\ \text{m} \quad (\approx 100\ \text{Mpc})$$

$$H_0 = \frac{v}{d} = \frac{7.13 \times 10^{6}}{3.15 \times 10^{24}} = 2.26 \times 10^{-18}\ \text{s}^{-1}$$

$$t \approx \frac{1}{H_0} = 4.42 \times 10^{17}\ \text{s} = \frac{4.42 \times 10^{17}}{3.15 \times 10^{7}} = 1.4 \times 10^{10}\ \text{years}$$

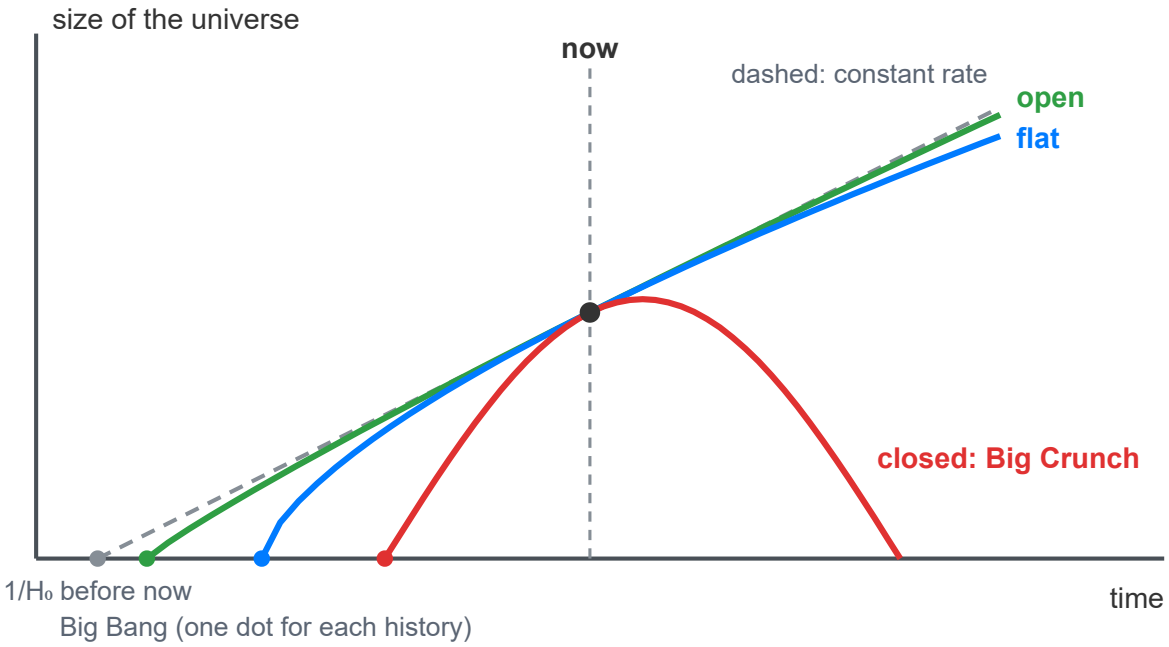
Why the value of H_0 is controversial

Redshifts are easy to measure accurately; distances are not. For most of the 20th century different teams found values of H_0 between about 50 and 100 km s⁻¹ Mpc⁻¹, so the age $1/H_0$ could be anywhere from about 10 to 20 billion years. At the high end the universe would be younger than the oldest stars in it, which is impossible. Today's measurements are much better, but small disagreements remain.

Dark matter and the fate of the universe

Gravity pulls every galaxy towards every other one, so it slows the expansion down. What happens in the end depends on how much mass there is, or more precisely the average **density** of the universe:

- **Low density (open universe):** gravity is too weak to stop the expansion, which continues for ever.
- **Critical density (flat universe):** the expansion slows down, getting closer and closer to stopping but never quite reversing.
- **High density (closed universe):** the expansion stops and reverses, and everything collapses back together in a "Big Crunch".



Three possible histories. All of them match the expansion rate we measure today (the same gradient at "now"). A slowing universe started more recently than $1/H_0$ ago, and the denser it is, the younger it is. The dots on the time axis mark where each one began.

The missing mass: dark matter

To predict the fate we need the density, but most of the mass in the universe seems to be invisible. One strong piece of evidence comes from how galaxies rotate. Stars orbit the centre of a galaxy just as planets orbit the Sun, so their speed should obey $v = \sqrt{GM/r}$, where M is the mass inside their orbit (see Gravitational Fields). Most

of the visible stars are concentrated near the centre, so stars far out should move more slowly, like the outer planets. They don't: the orbital speed stays roughly **constant** far beyond the visible edge of the galaxy. There must be extra, unseen mass spread through a huge halo around it.

This **dark matter** does not emit or absorb light; we only detect it through its gravity. Measurements suggest there is roughly five times as much of it as ordinary matter, but nobody yet knows what it is.

KEY IDEA

The controversy in a nutshell

- **Age:** $t \approx 1/H_0$, so the age is only as good as our value of H_0, and that depends on difficult distance measurements. If the expansion has slowed, the true age is less than $1/H_0$.
- **Fate:** it depends on the density of the universe compared with a critical density, and the critical density itself depends on H_0.
- **Dark matter:** most of the mass is invisible and of unknown nature, so the density, and therefore the fate, is hard to pin down.

Quick checks

QUICK CHECK

Star X has the same radius as star Y but twice the surface temperature. What is the luminosity of X divided by the luminosity of Y?

- 2
- 4
- 8
- 16 ✓

$L = \sigma 4\pi r^2 T^4$ with the same r, so $L \propto T^4 = 2^4 = 16$.

QUICK CHECK

The spectrum of star P peaks at 300 nm and that of star Q peaks at 600 nm. Which statement is correct?

- P is cooler than Q and looks redder.
- P has twice the surface temperature of Q.
- P has half the surface temperature of Q.
- P must be more luminous than Q.

Wien's law: $\lambda_{\max} \propto 1/T$, so half the peak wavelength means twice the temperature. You can't say which is more luminous without knowing their radii.

QUICK CHECK

A star appears in the top right-hand corner of a Hertzsprung–Russell diagram. What can you say about it?

- It is hot and small.
- It is hot and large.
- It is cool and very large.
- It is cool and small.

The temperature axis runs from hot on the left to cool on the right, so the top right means low T but high L. From $L = 4\pi r^2 \sigma T^4$, that needs a huge radius: a red giant or supergiant.

QUICK CHECK

Star A has a parallax angle of 0.05 arcseconds and star B has a parallax angle of 0.10 arcseconds. Which statement is correct?

- [x] A is twice as far away as B.
- [ ] B is twice as far away as A.
- [ ] A is four times as far away as B.
- [ ] They are the same distance away.

$d = 1/p$: A is at 20 pc and B at 10 pc. A smaller parallax angle means a more distant star.

QUICK CHECK

A spectral line at 486 nm in the laboratory is observed at 491 nm in light from a galaxy. What is the galaxy's velocity?

- [ ] $3.1 \times 10^{6}\ \mathrm{m\ s^{-1}}$ towards us
- [x] $3.1 \times 10^{6}\ \mathrm{m\ s^{-1}}$ away from us
- [ ] $3.1 \times 10^{5}\ \mathrm{m\ s^{-1}}$ away from us
- [ ] $3.0 \times 10^{8}\ \mathrm{m\ s^{-1}}$ away from us

$z = 5/486 = 0.0103$, so $v = zc = 3.1 \times 10^{6}\ \mathrm{m\ s^{-1}}$. The wavelength got longer (redshift), so the galaxy is receding.

QUICK CHECK

A new measurement gives a larger value of the Hubble constant than before. What does this suggest about the universe?

- It is older than previously thought.
- **It is younger than previously thought.**
- Its age is unchanged; only distances change.
- It must be closed and will end in a Big Crunch.

The age is about $1/H_0$, so a bigger H_0 means a younger universe: galaxies are moving apart faster, so it took less time to reach their present separations.